

Automatic Image Analysis

Transformers II

Simon Prince: Understanding Deep Learning, Chapter 12

Julia Turc: <https://www.youtube.com/watch?v=LE3NfEULV6k>

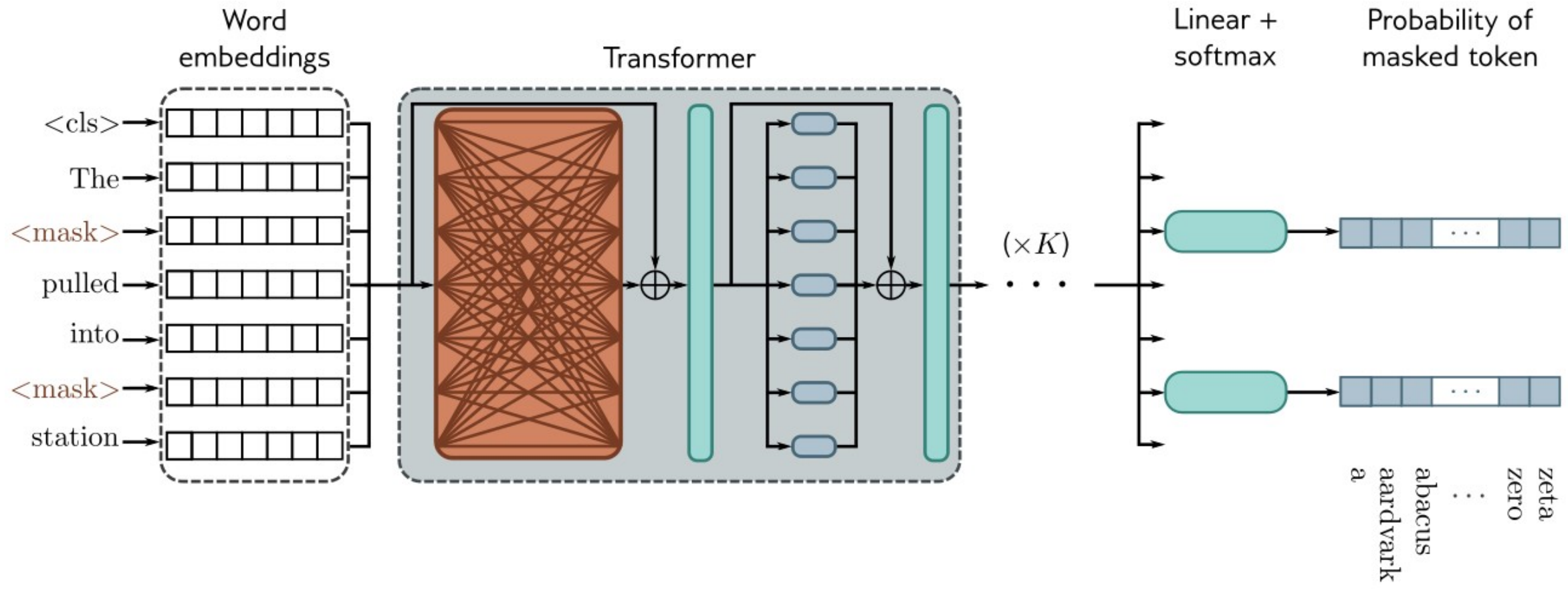
<https://www.youtube.com/watch?v=jNPmSjrxGtY>

Transformer Model

- The embedding matrix **X** is passed through a series of K transformer layers.
- **Encoder** transforms the text embeddings into a **representation** that can **support a variety of tasks** (e.g. text classification. BERT).
- **Decoder** predicts next token to continue the input. (e.g. completing test; story telling. GPT-3)
- **Encoder-decoders** are used in sequence to sequence tasks (e.g. machine translation)

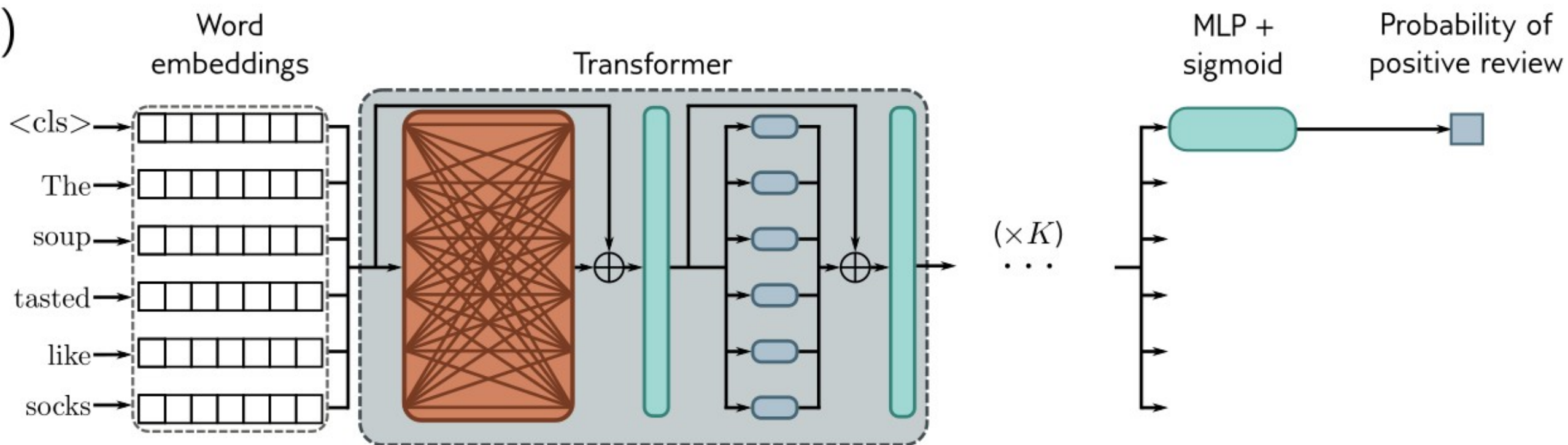
Encoder Pre-Training

- 1) Input tokens are converted to word embeddings $\mathbf{x}^T$.
- 2) Input is passed through transformer layers to create output embeddings.
- 3) In pre-training, the goal is to predict masked embeddings.
- 4) Output embeddings are passed through softmax function predicting probabilities of missing words w.r.t. the vocabulary.



Encoder Fine-Training

- **Text classification**
- Fine-tuning with manually labelled data
- Linear transformation or multi-layer perceptron
- Here: sentiment classification



Transfer Learning

Pre-training: Language modeling



WIKIPEDIA
The Free Encyclopedia

King is the title given to a male ---

monarch ██████████
body ██████
dog ████
symbol █

Fine-tuning: Sentiment Classification

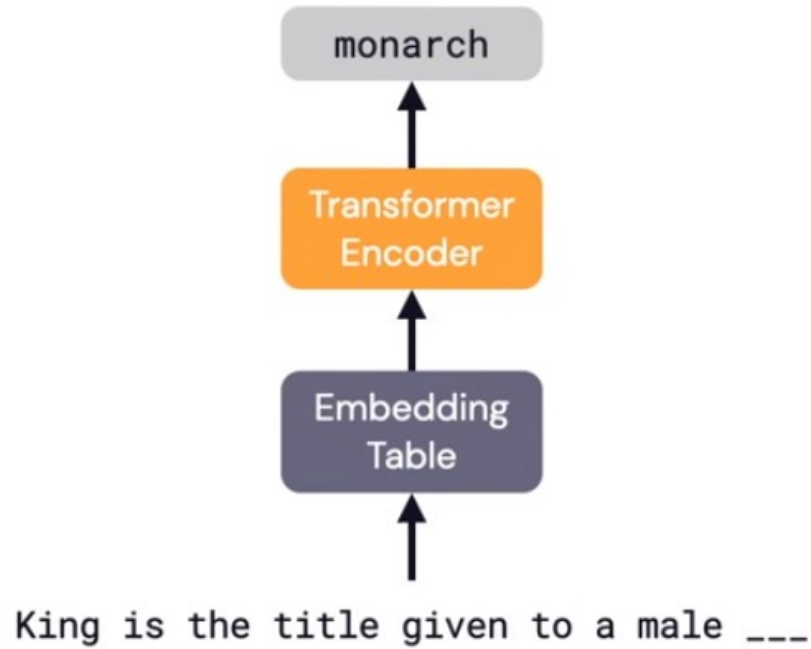


Bittersweet and brilliant to the very end, BoJack Horseman's final season manages to keep surprising viewers with its empathy and depth.

positive ██████████
negative █

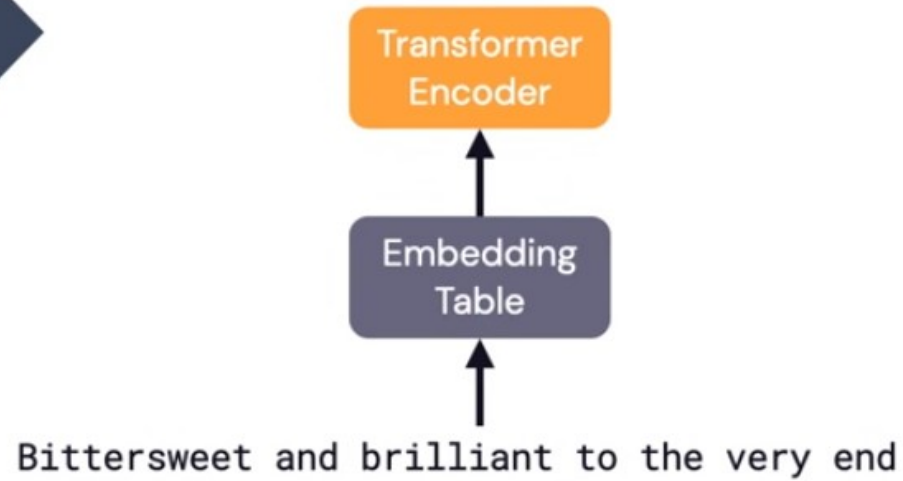
Transfer Learning with Transformers

Pre-training: Language modeling



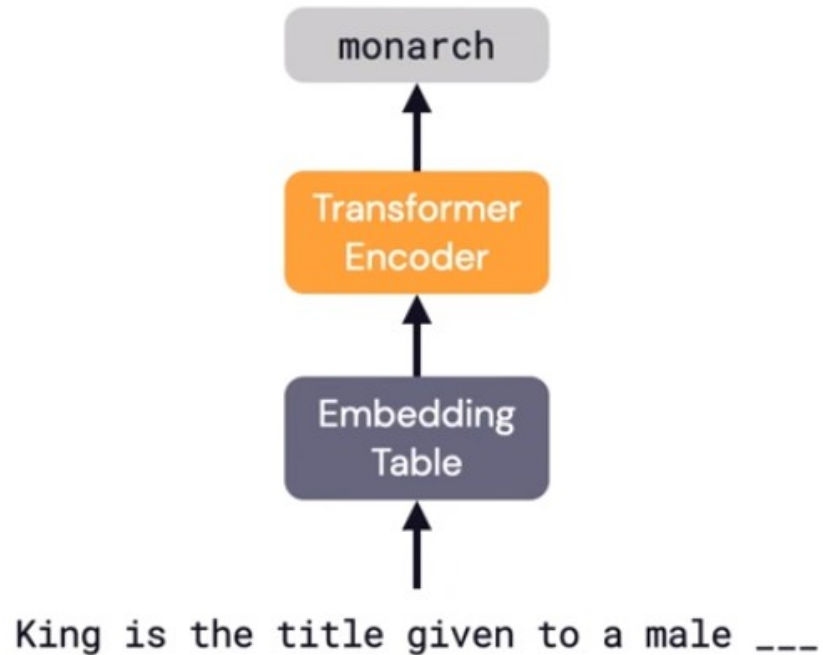
Copy

Fine-tuning: Sentiment Classification



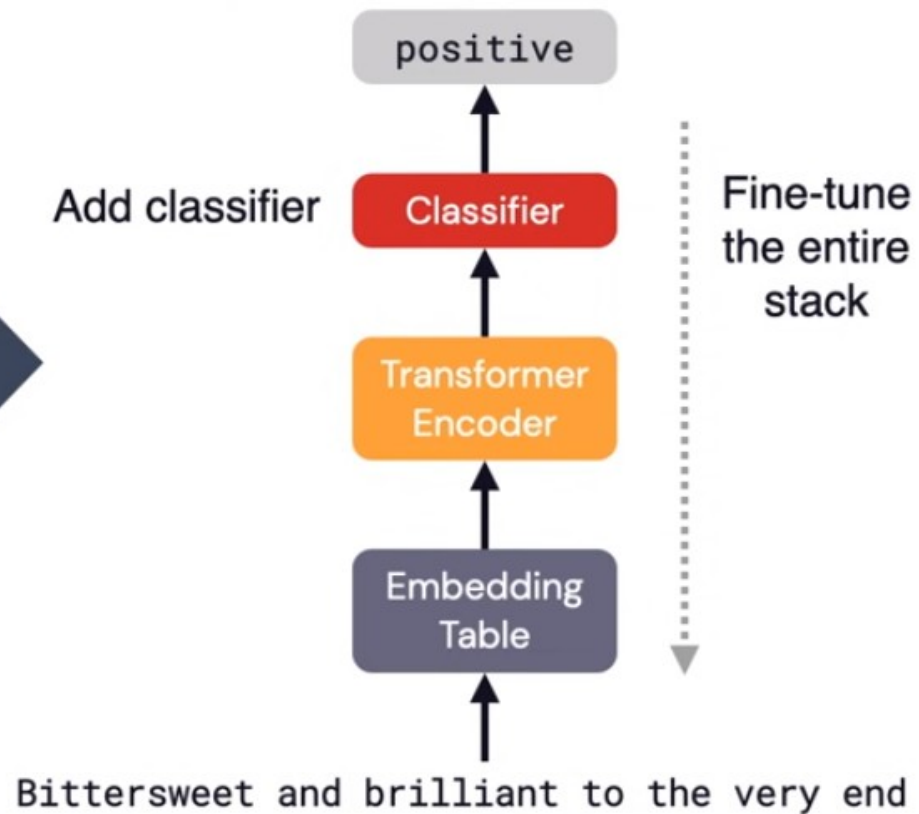
Transfer Learning with Transformers

Pre-training: Language modeling



Copy

Fine-tuning: Sentiment Classification



BERT: Bidirectional Encoder Representations from Transformers

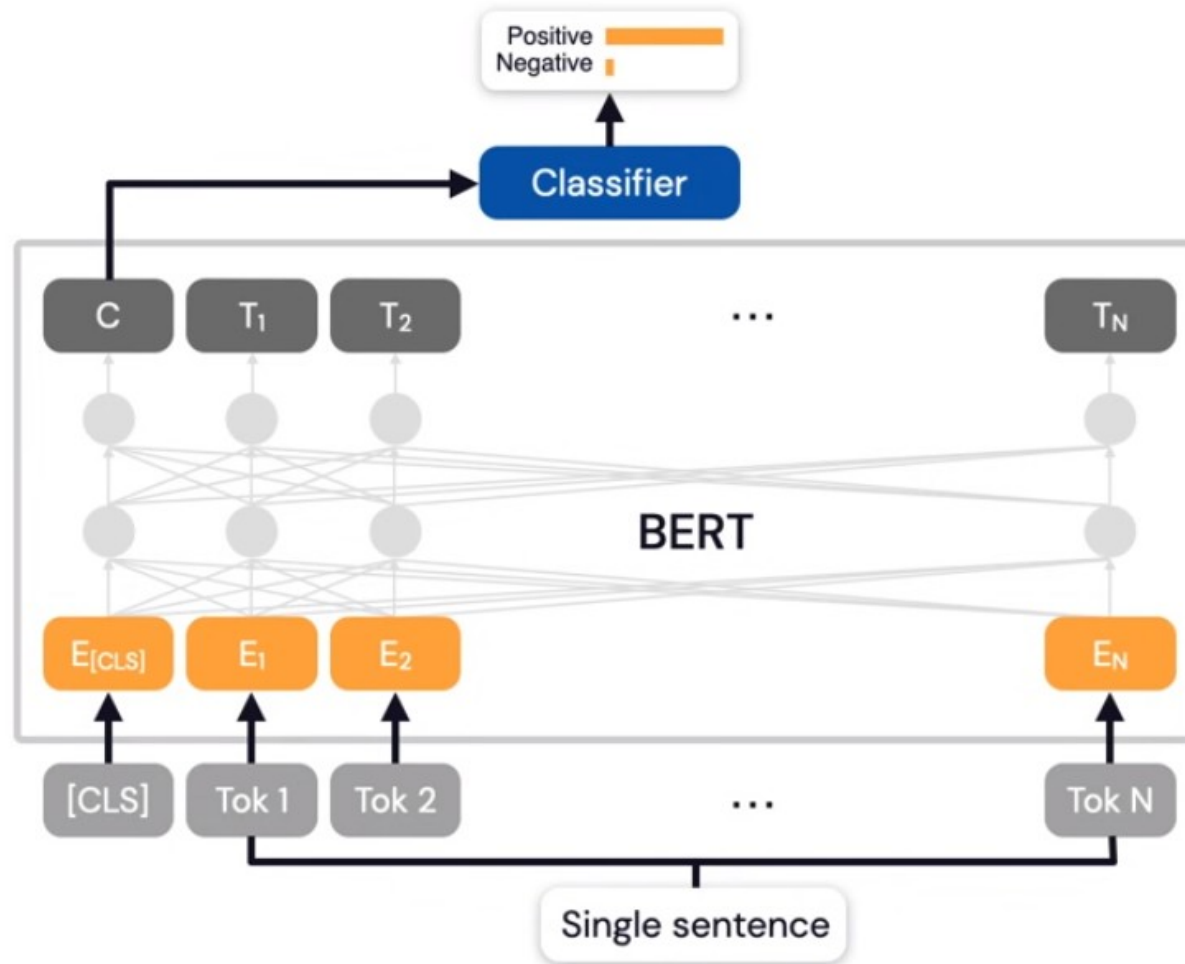
- BERT is a Transformer Encoder trained on Wikipedia
- Readily available for download on:
 - English (24 different sizes)
 - Chinese
 - Multilingual (104 languages)

Finetuning BERT for Sentence Classification

e.g. sentiment classification

Bittersweet and brilliant to the very end, BoJack Horseman's final season manages to keep surprising viewers with its empathy and depth.

Positive 
Negative 

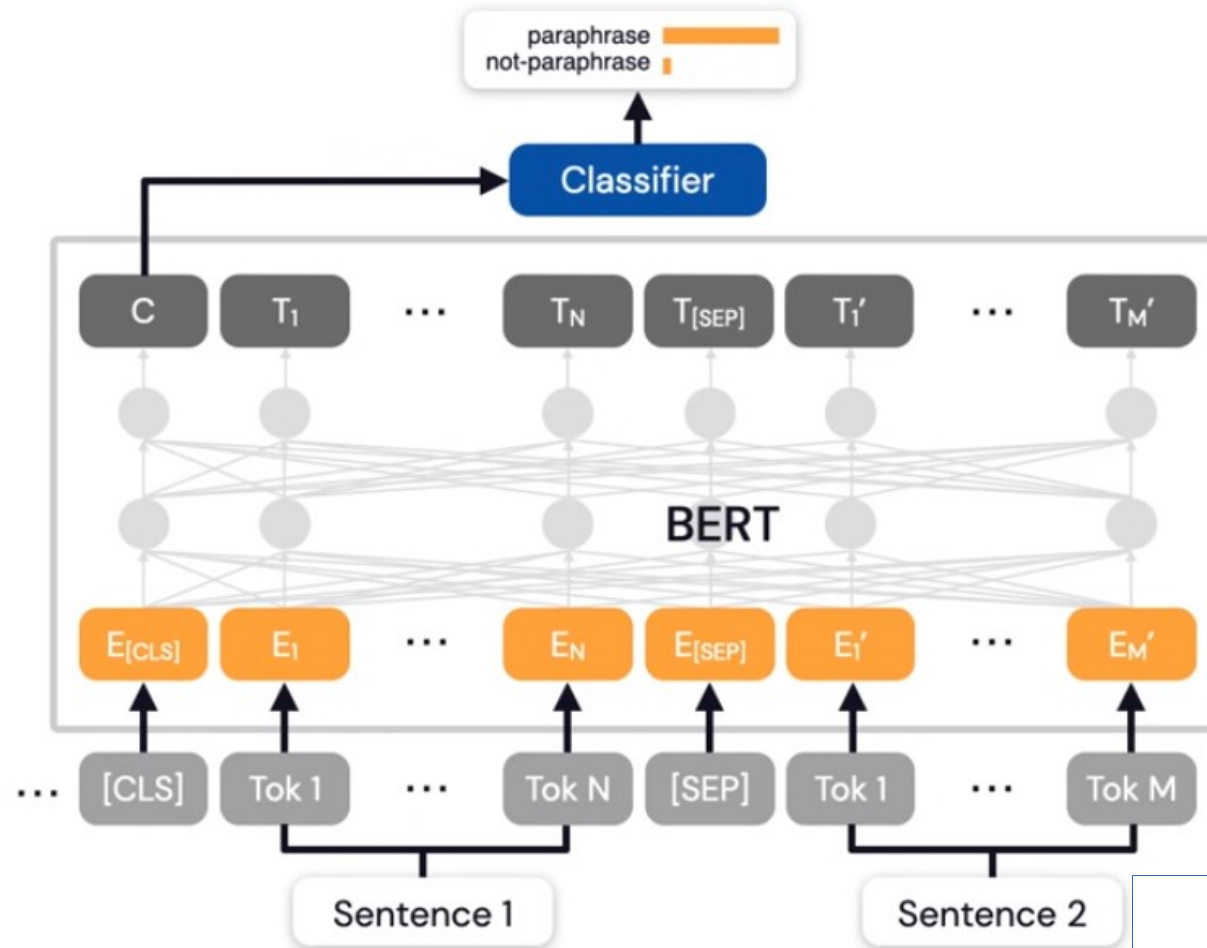


Finetuning BERT for Sentence Pair Classification

e.g. paraphrase classification

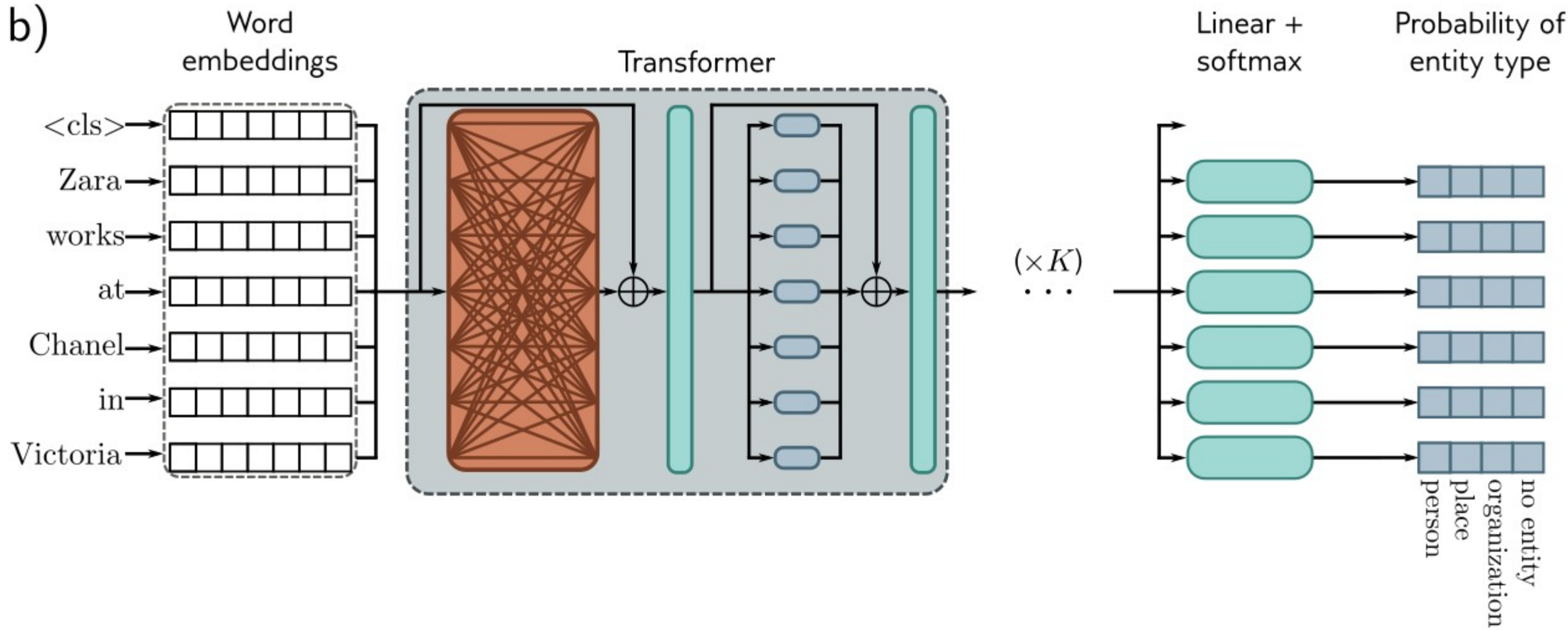
1. She accepted immediately
2. The woman did not hesitate to accept

paraphrase 
not-paraphrase 



Encoder Fine-Training

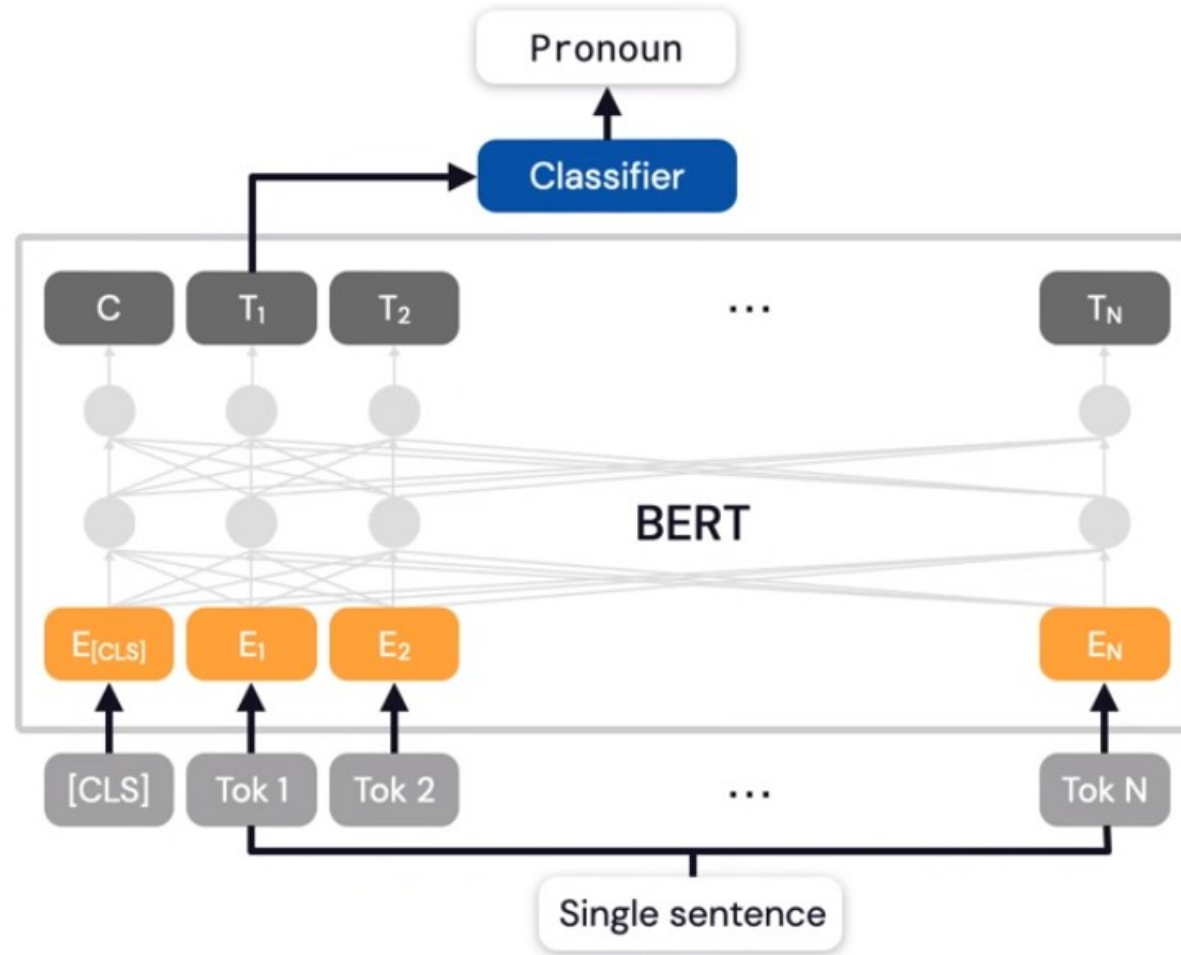
- **Word classification**
- Named entity recognition: person, place, organization



Finetuning BERT for Span Annotation

e.g. part of speech tagging

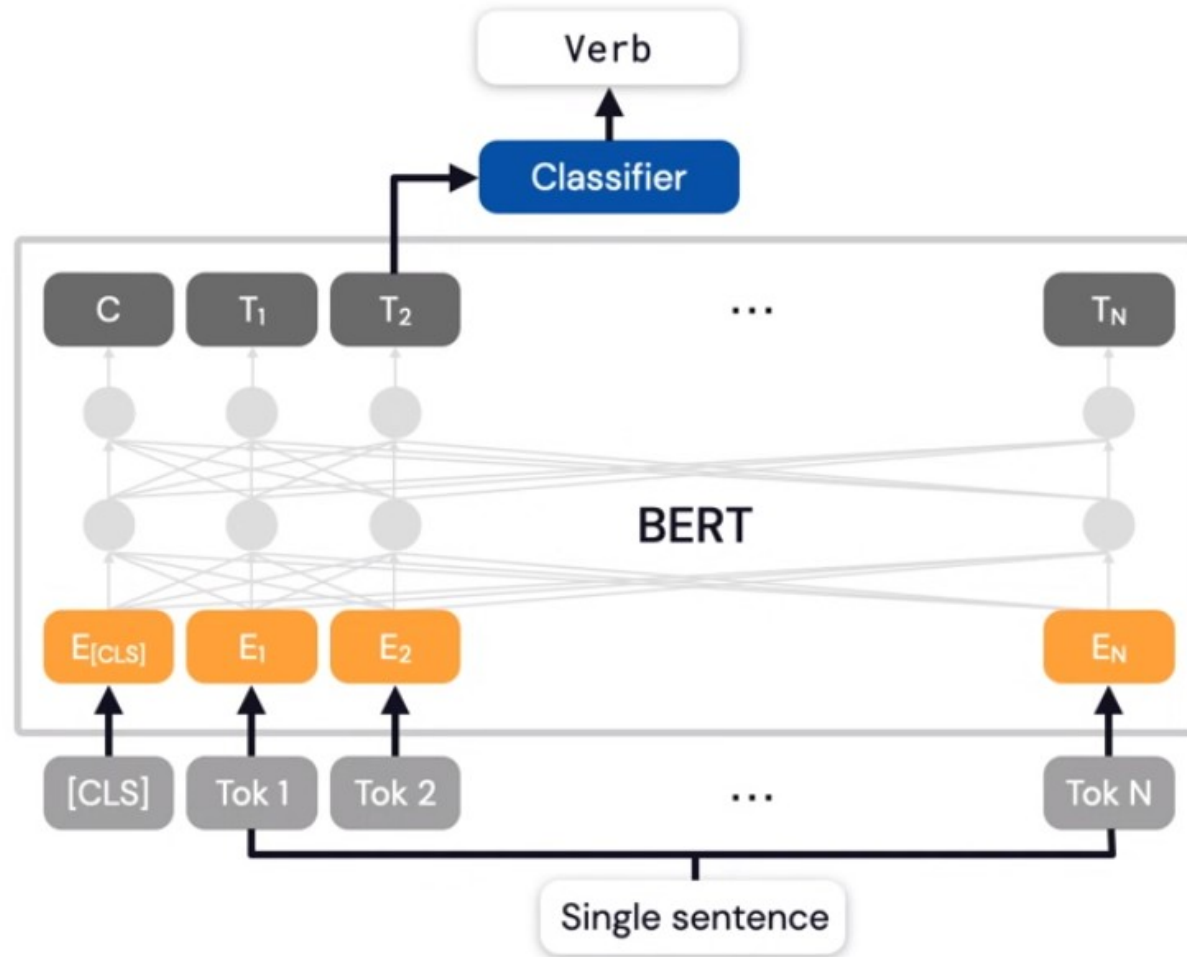
She accepted immediately
| | |
| | |
| Verb Adverb
Pronoun



Finetuning BERT for Span Annotation

e.g. part of speech tagging

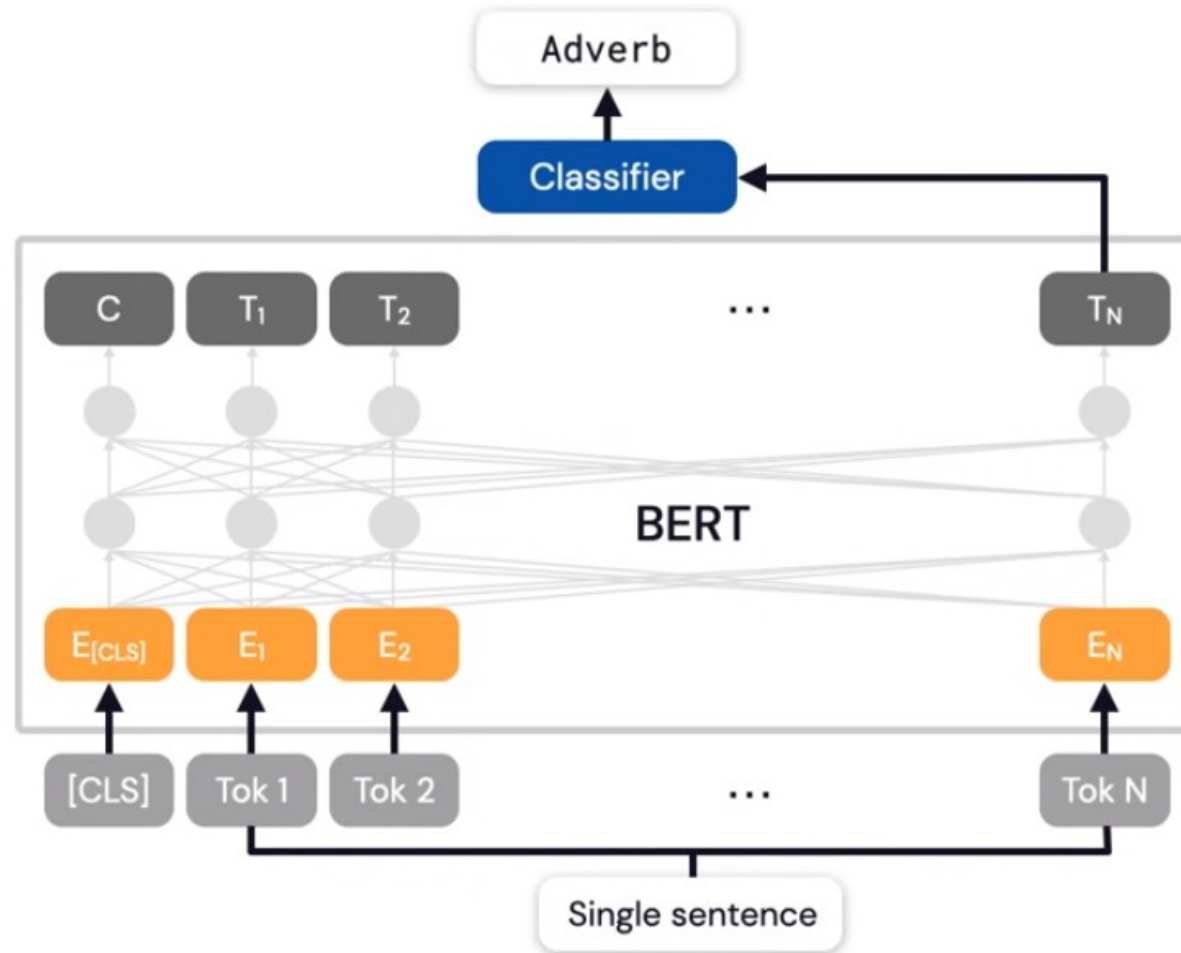
She accepted immediately
| | |
| | |
| Verb Adverb
Pronoun



Finetuning BERT for Span Annotation

e.g. part of speech tagging

She accepted immediately
| | |
| | |
| Verb Adverb
Pronoun



Using BERT via TF-Hub

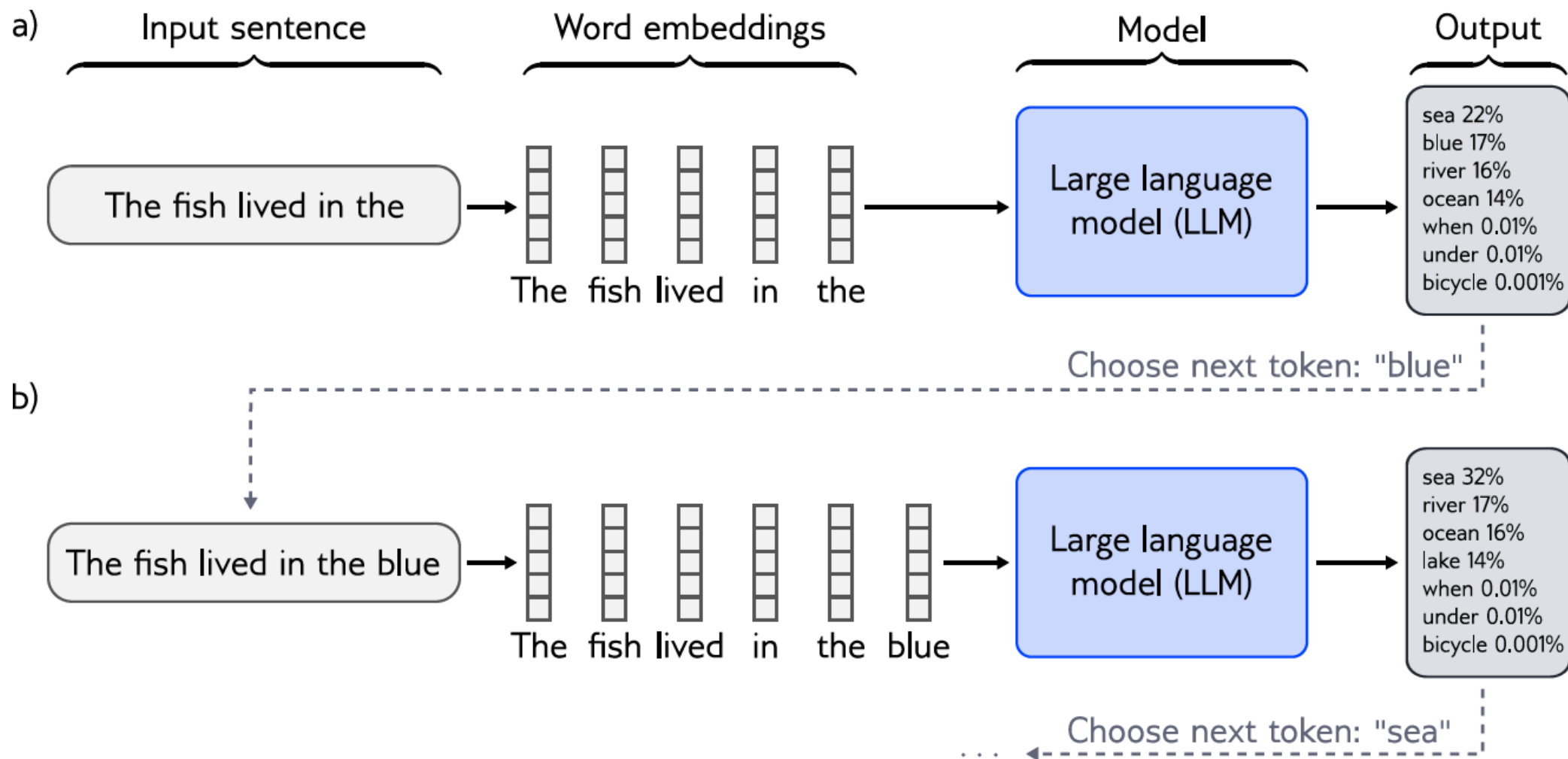
```
1 seq_length = 128
2
3 # Get a small BERT model from TF-Hub with sequence length 128
4 bert = hub.KerasLayer (
5     "https://tfhub.dev/tensorflow/small_bert/bert_en_uncased_L-2_H-128_A-2/1",
6     trainable=True)
7
8 # Define the inputs.
9 inputs = dict(
10     input_word_ids=tf.keras.layers.Input (shape=(seq_length,), dtype=tf.int32),
11     input_mask=tf.keras.layers.Input(shape=(seq_length,), dtype=tf.int32),
12     input_type_ids=tf.keras.layers.Input(shape=(seq_length,), dtype=tf.int32))
13
14 # Get the embedding of the CLS token.
15 bert_outputs = bert(inputs)
16 cls_embedding = bert_outputs["pooled_output"]
17
18 # Define a classifier for sentiment classification.
19 classifier = tf.keras.layers.Dense(units=2, activation="softmax")
20 classifier_output = classifier(cls_embedding)
21
22 # Put everything together.
23 model = tf.keras.Model(inputs=inputs, outputs=classifier_output)
```

The BERTology

BERT's success inspired a lot of follow-up work:

- RoBERTa (more robust training)
- More efficient models (fewer parameters, faster inference):
 - ALBERT
 - MobileBERT
 - TinyBERT...
- Language-specific models:
 - French: FlauBERT, CamemBERT
 - Chinese: AMBERT
 - ...

Decoder model



Decoder model: GPT3

- GPT: Generative Pre-trained Transformer
- One job: predict the next word in a sequence
- More formally builds an **autoregressive** probability model

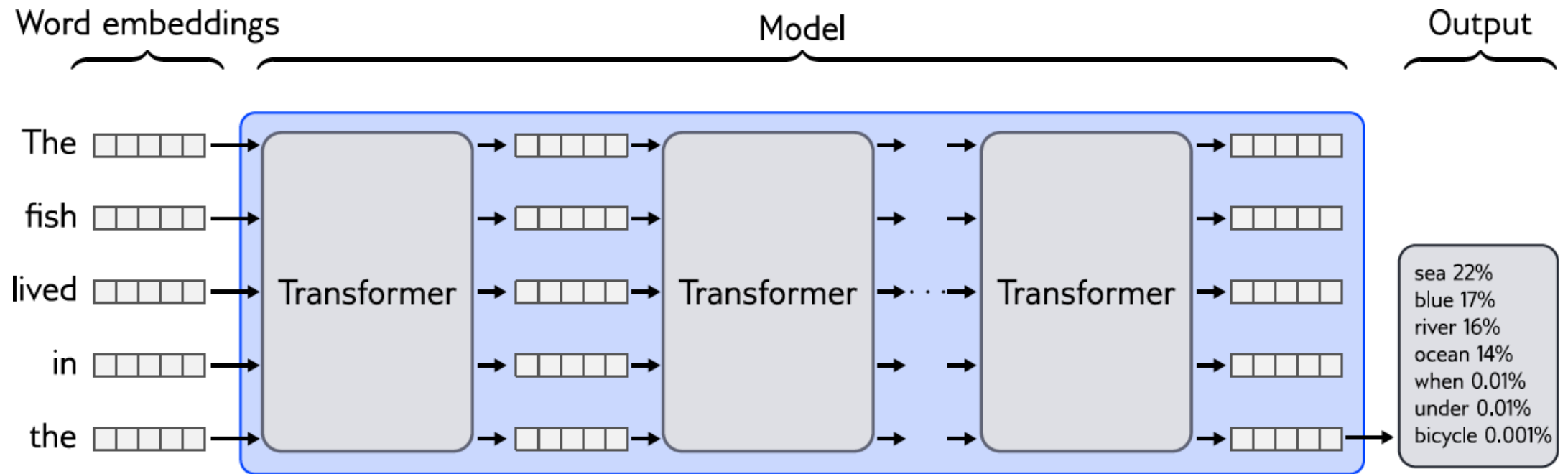
$$Pr(t_1, t_2, \dots t_N) = Pr(t_1) \prod_{n=2}^N Pr(t_n | t_1 \dots t_{n-1})$$

Decoder model: GPT3

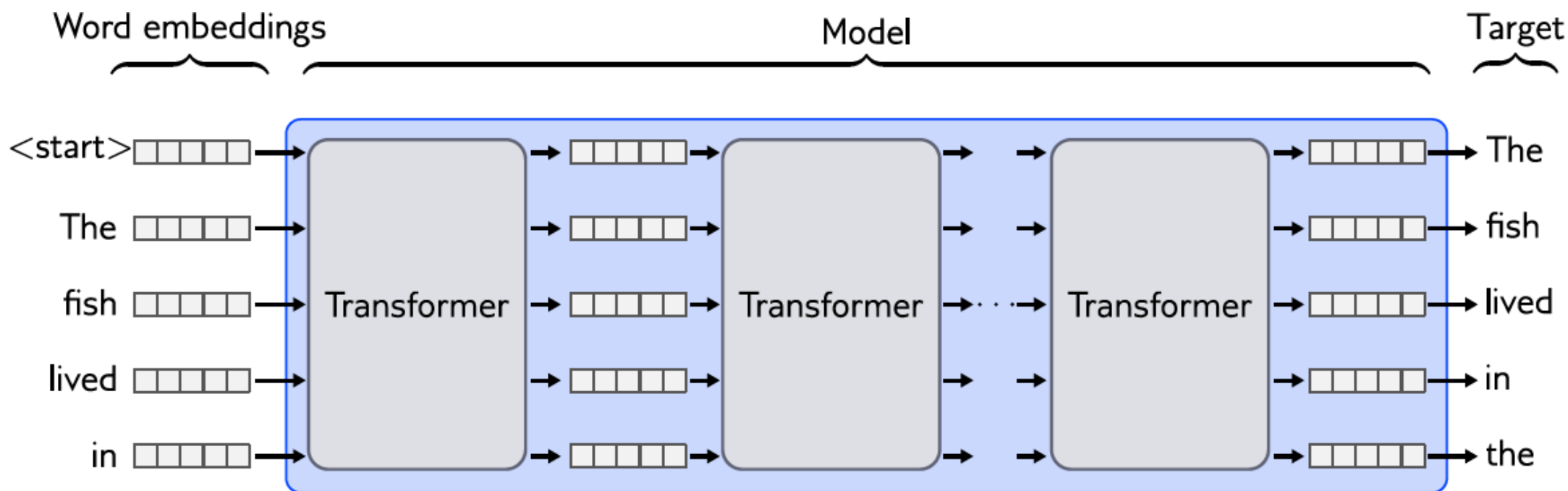
- Builds autoregressive probability model
- E.g. “It takes great courage to let yourself appear weak”

$$\begin{aligned} Pr(\text{It takes great personal courage to let yourself appear weak}) = \\ Pr(\text{It}) \times Pr(\text{takes}|\text{It}) \times Pr(\text{great}|\text{It takes}) \times Pr(\text{courage}|\text{It takes great}) \times \\ Pr(\text{to}|\text{It takes great courage}) \times Pr(\text{let}|\text{It takes great courage to}) \times \\ Pr(\text{yourself}|\text{It takes great courage to let}) \times \\ Pr(\text{appear}|\text{It takes great courage to let yourself}) \times \\ Pr(\text{weak}|\text{It takes great courage to let yourself appear}). \end{aligned}$$

Predicting next



Predicting all next words simultaneously



Few shot learning:

Poor English input: I eated the purple berries.

Good English output: I ate the purple berries.

Poor English input: Thank you for picking me as your designer. I'd appreciate it.

Good English output: Thank you for choosing me as your designer. I appreciate it.

Poor English input: The mentioned changes have done. or I did the alteration that you requested. or I changed things you wanted and did the modifications.

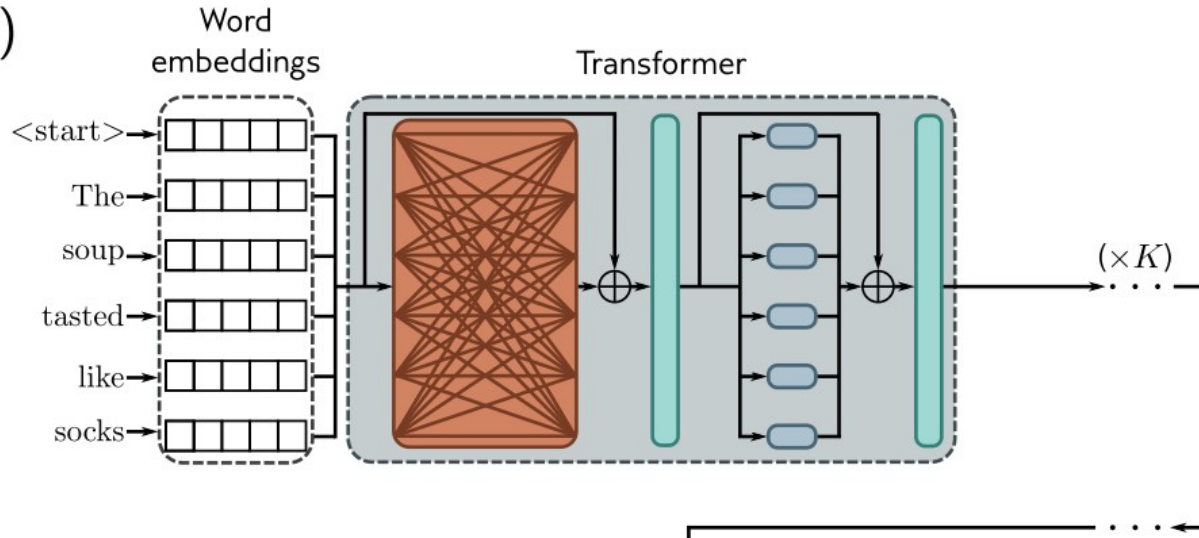
Good English output: The requested changes have been made. or I made the alteration that you requested. or I changed things you wanted and made the modifications.

Poor English input: I'd be more than happy to work with you in another project.

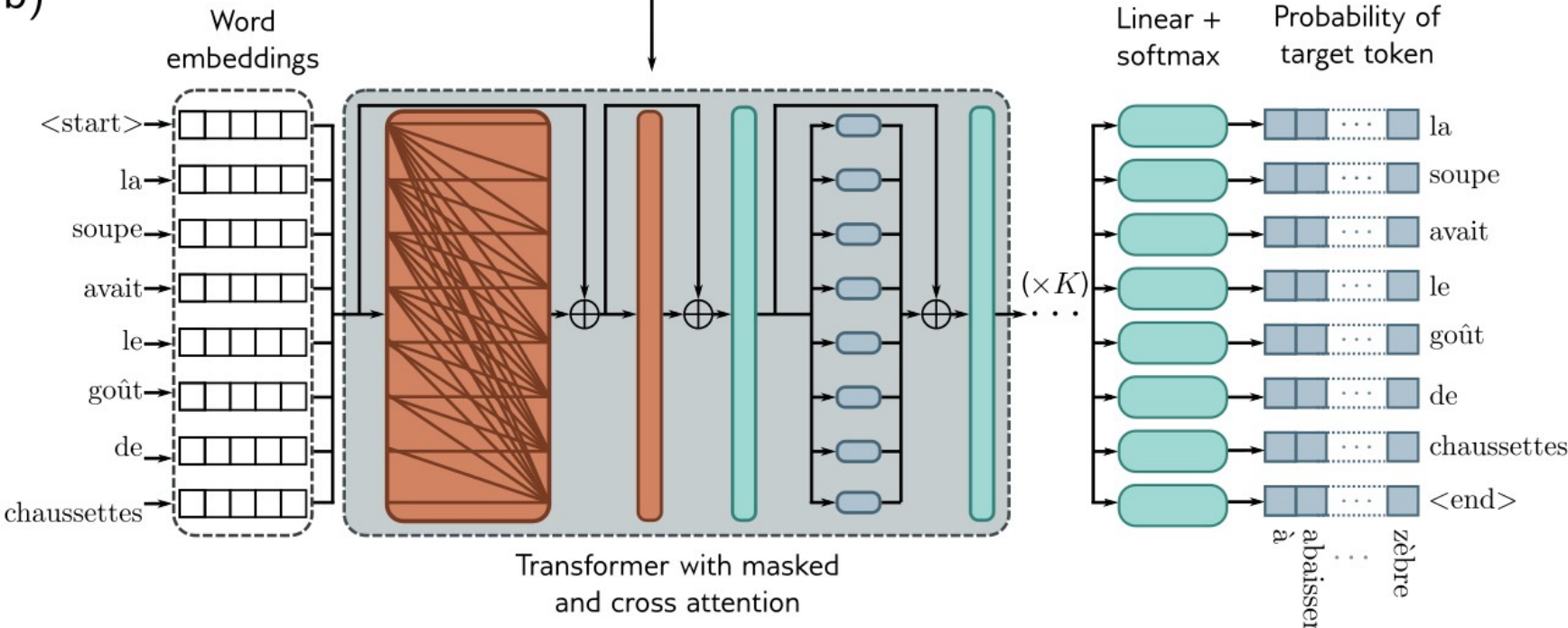
Good English output: I'd be more than happy to work with you on another project.

Encoder-Decoder for Machine Translation

a)



b)



- **Goal: translate the first into the second sentence.**

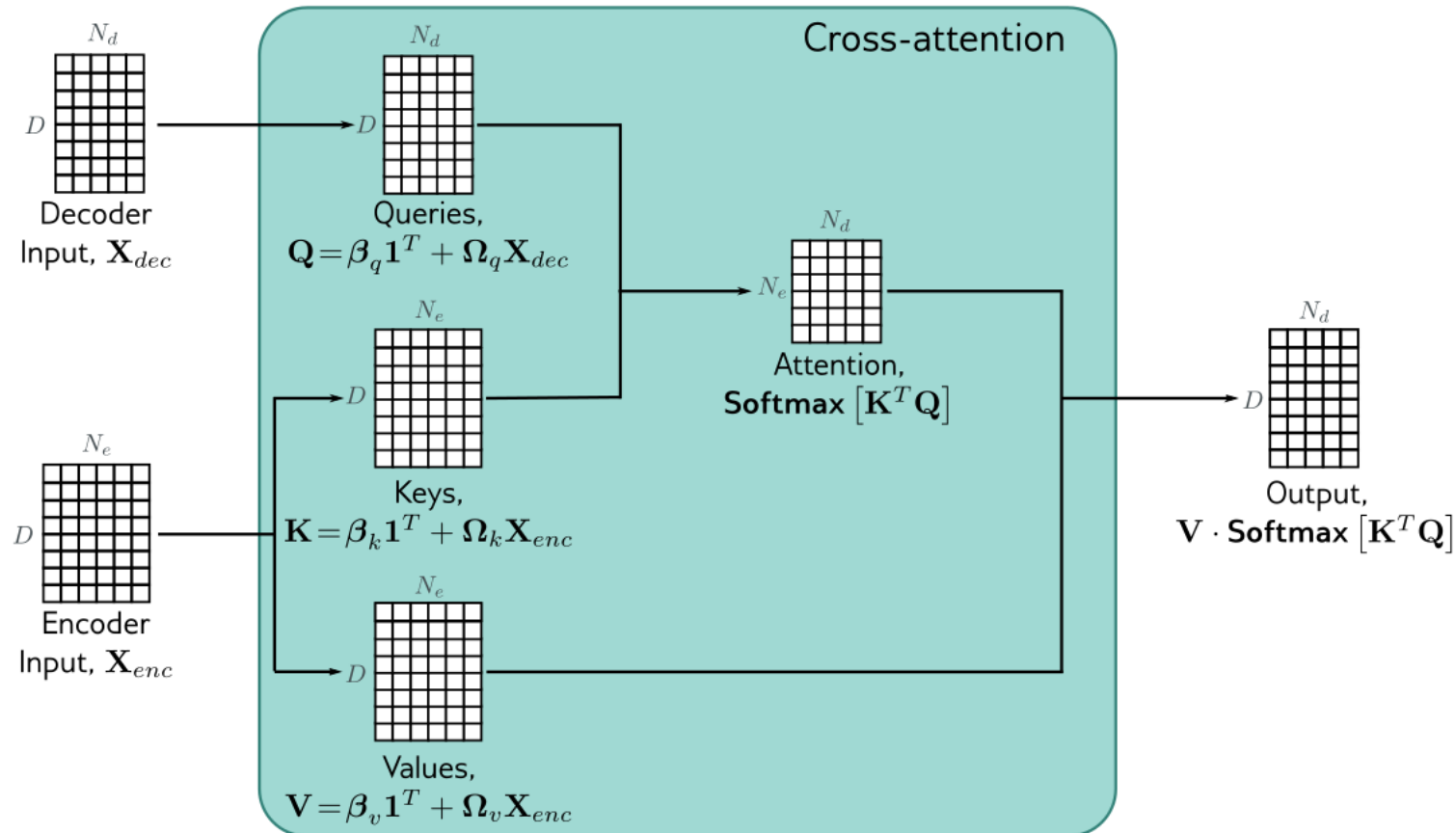
a) First sentence is passed to standard encoder.

b) Second sentence is passed through a decoder using masked self-attention, **cross-attending to output embeddings of the encoder**

- **Loss as for decoder: max. prob. of next word**

Cross-Attention

- Computation flow as in standard self-attention
- Queries calculated from decoder embeddings
- Keys and values calculated from encoder embeddings
- Decoder contains target language statistics



Transformers for Long Sequences

- In a transformer each token interacts with every other token.
- Computational complexity therefore scales quadratically.
- Interaction matrices for self-attention:

a) Encoder

b) Decoder

c) Convolutional encoder

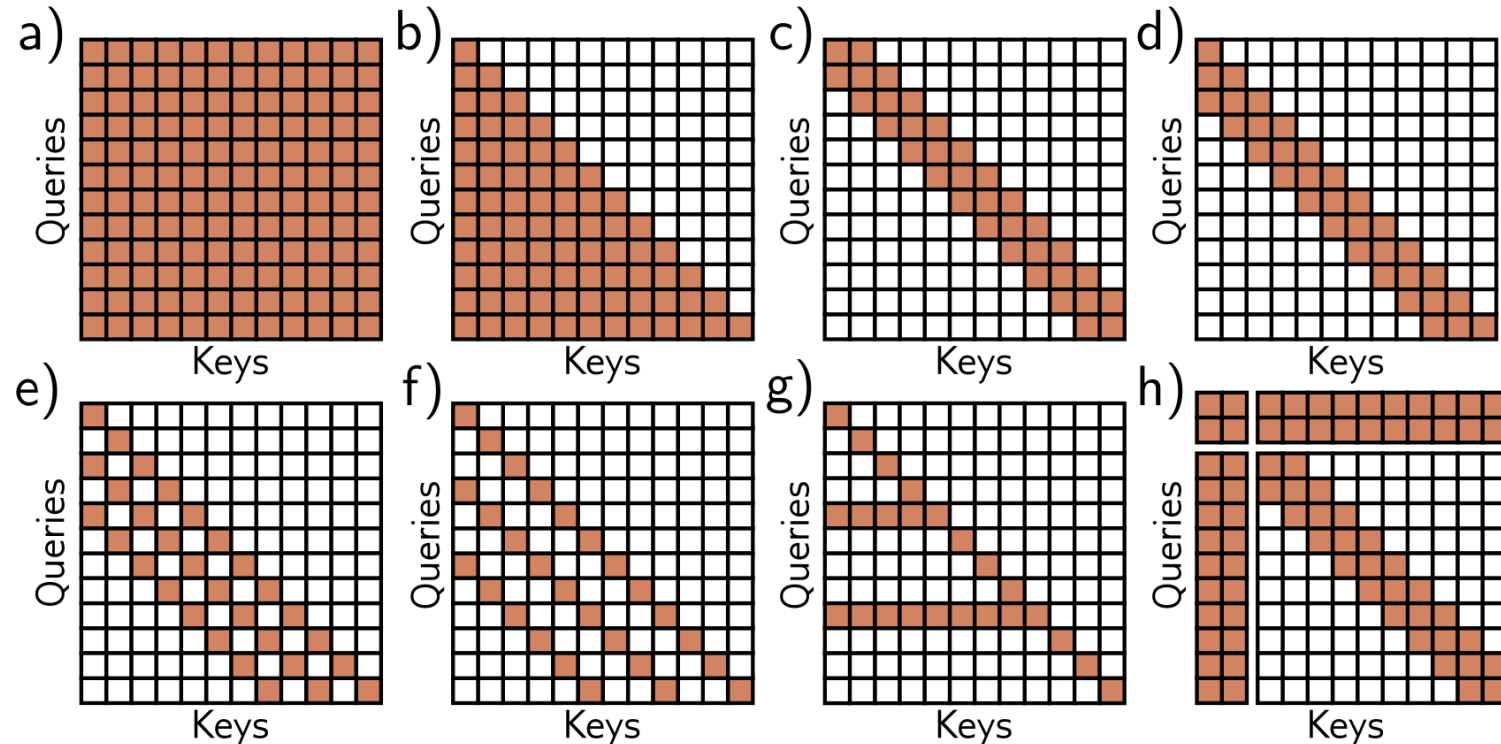
d) Convolutional decoder

e) Convolutional dilation 2

f) Convolutional dilation 3

g) Allow selected tokens to interact with all other

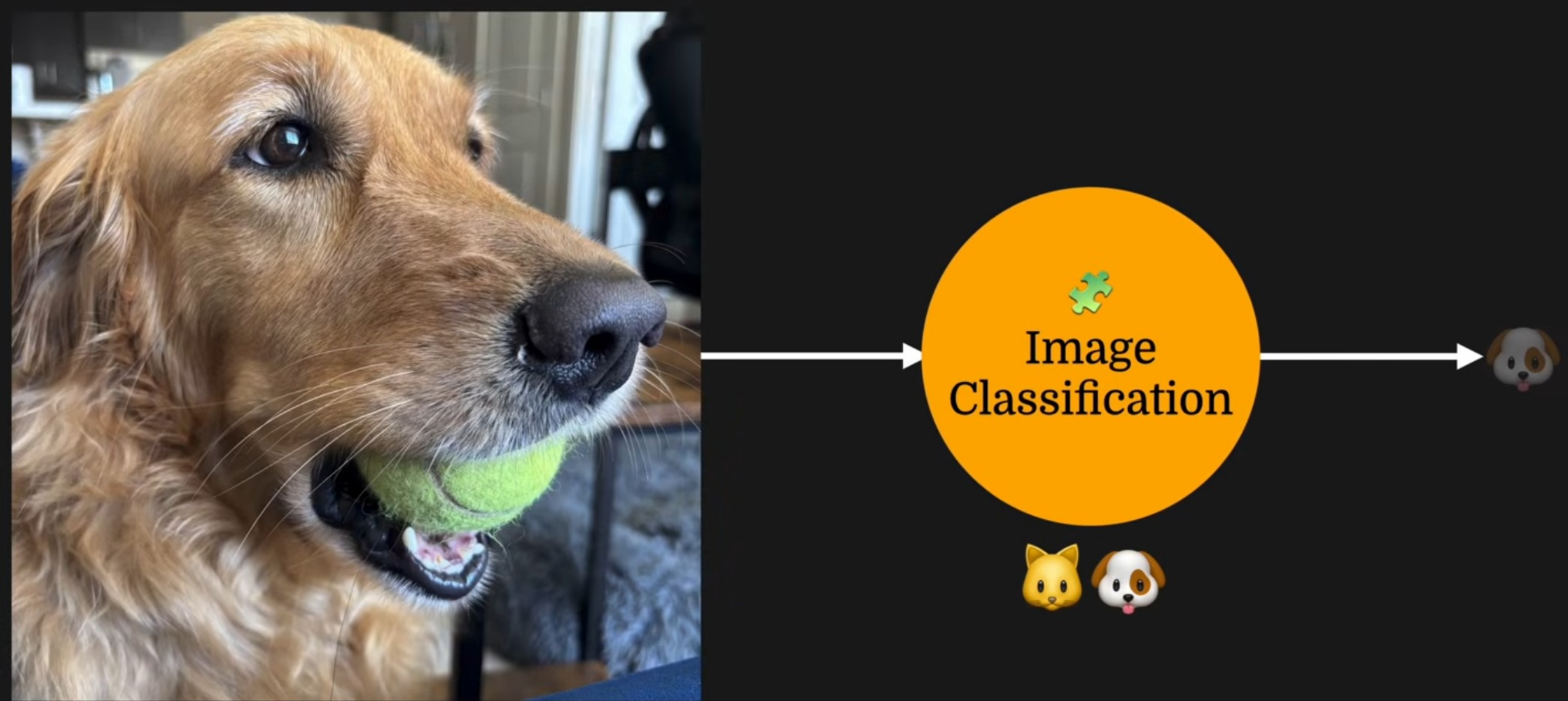
h) Global tokens interact with all tokens and with each other



Transformers for Images

- Not promising, as
 - there are any more pixels in an image than words in a sentence
 - the inductive bias (structurally given for convolutional nets)
 - to be equivariant to spatial translation
 - to take 2D structure of images into accountis not given for transformers.

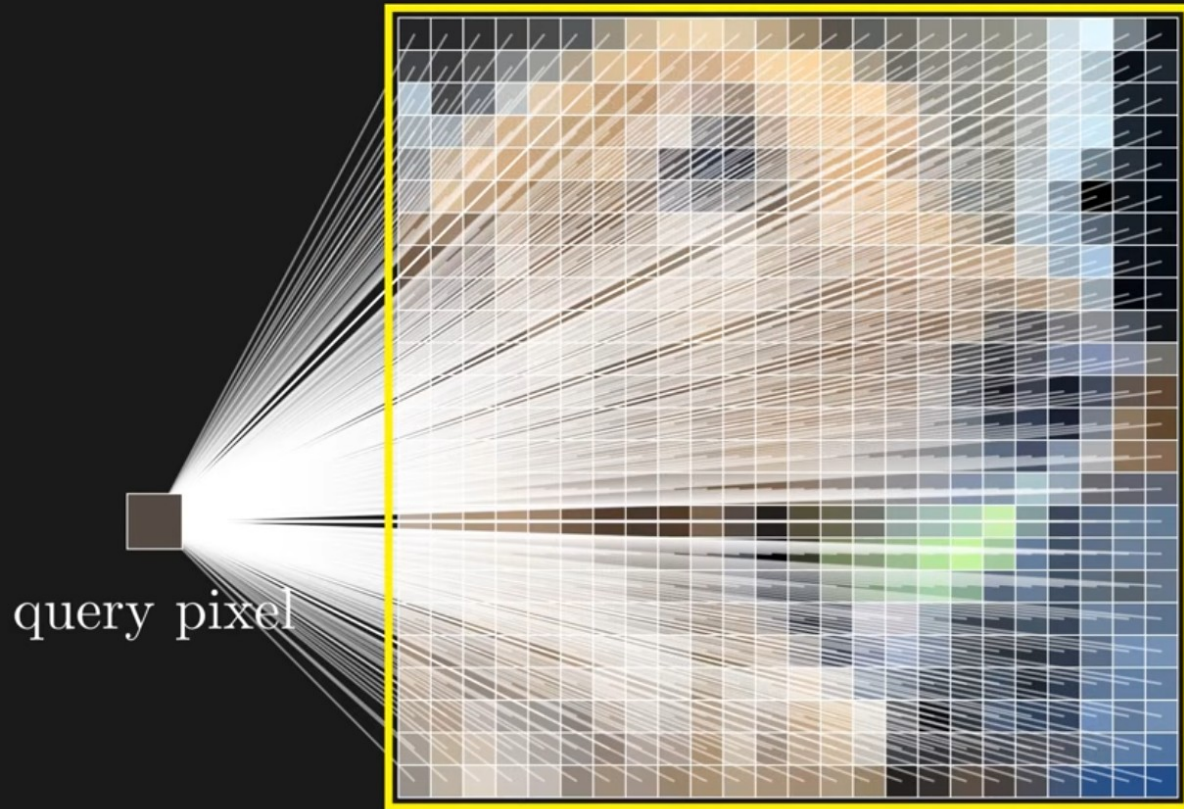
Image Classification Task



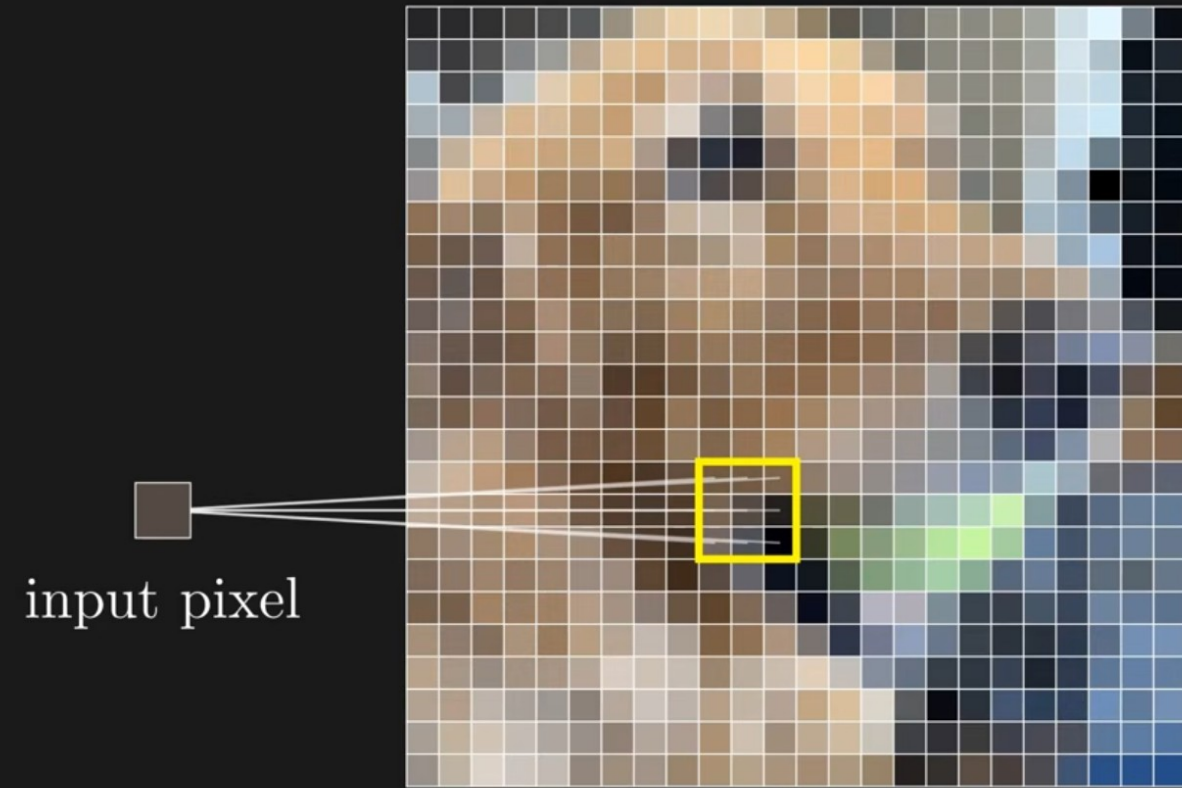
- Julia Turc: Diffusion Transformers (ViT, DiT, MMDiT),
<https://www.youtube.com/watch?v=jNPmSjrxGtY&t=359s>

Transformers vs. CNNs

Self-Attention



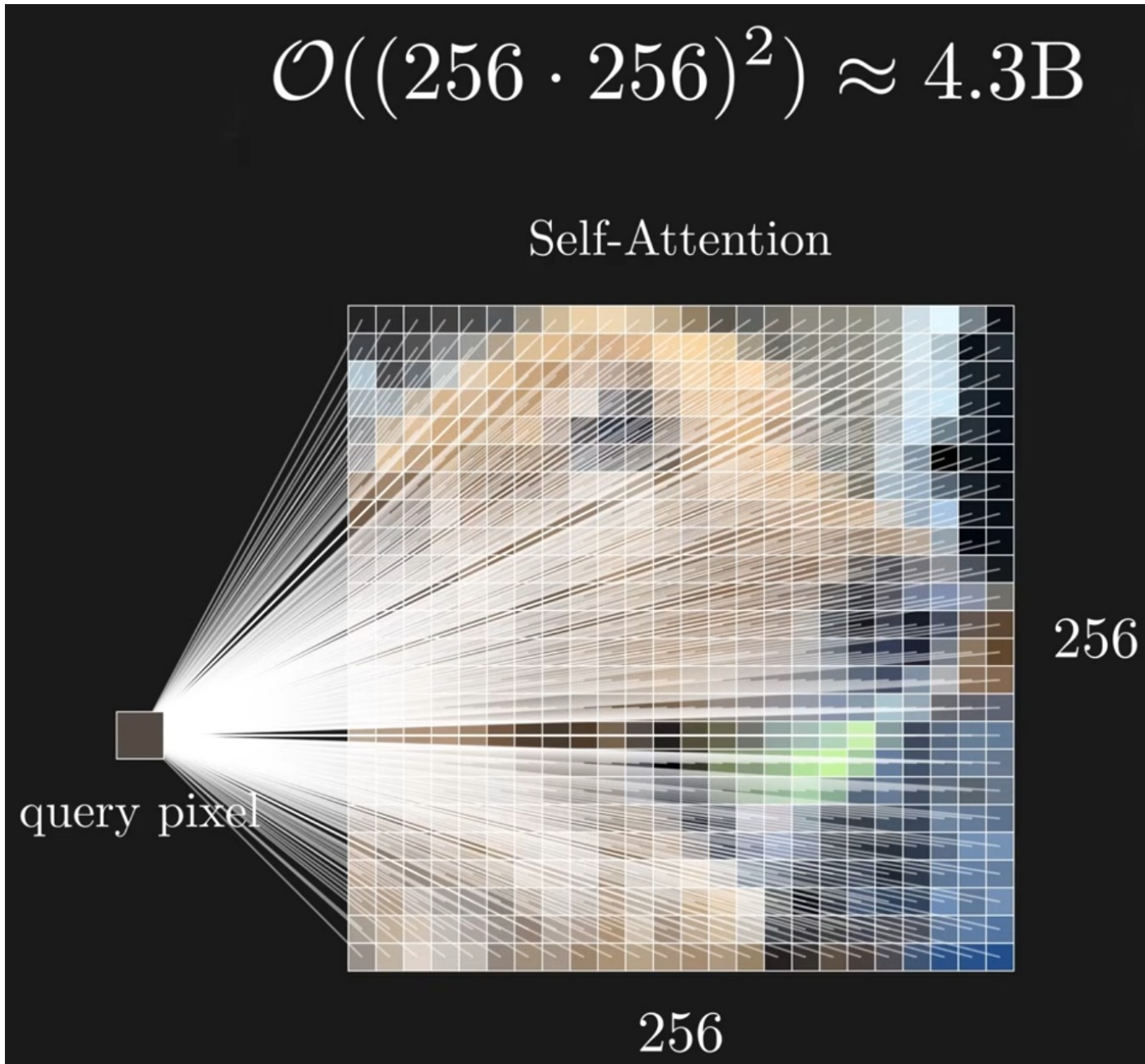
Convolution



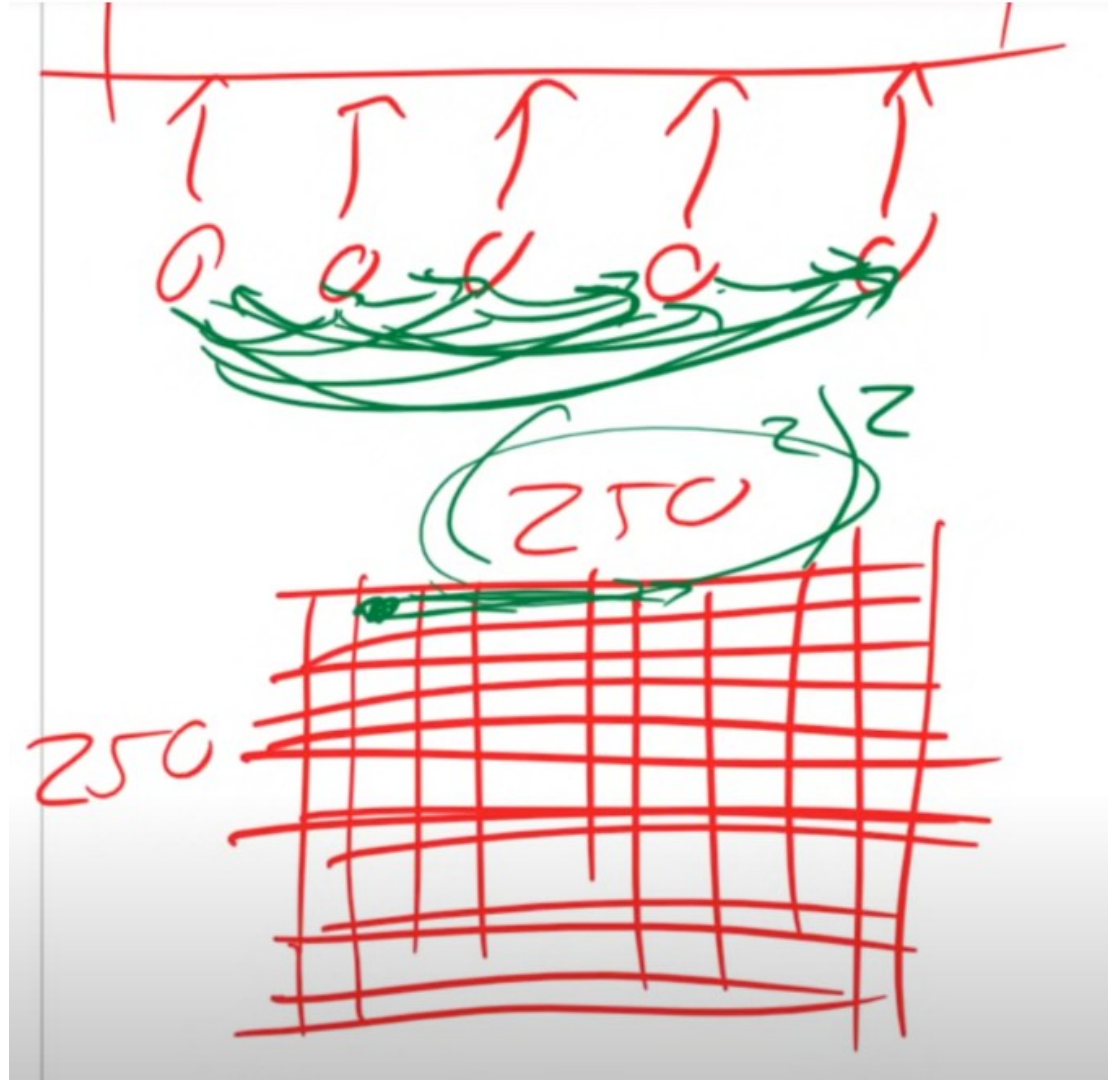
Transformers for Images

- Not promising, as
 - there are any more pixels in an image than words in a sentence
 - the inductive bias (structurally given for convolutional nets)
 - to be equivariant to spatial translation
 - to take 2D structure of images into accountis not given for transformers.
- Nevertheless, they have outperformed CNNs,
 - as large amounts of training data are available, and
 - as, then, learned inductive bias can be advantageous compared to network-architecturally given inductive bias.

Transformers for Images



Global Attention over Pixels



ImageGPT

- Transformer decoder
- Builds an autoregressive model of image pixel that
 - ingests a partial image and
 - predicts the subsequent pixel values
- Was able to operate on images of size 64x64 only
 - due to the quadratic complexity of transformers

ImageGPT

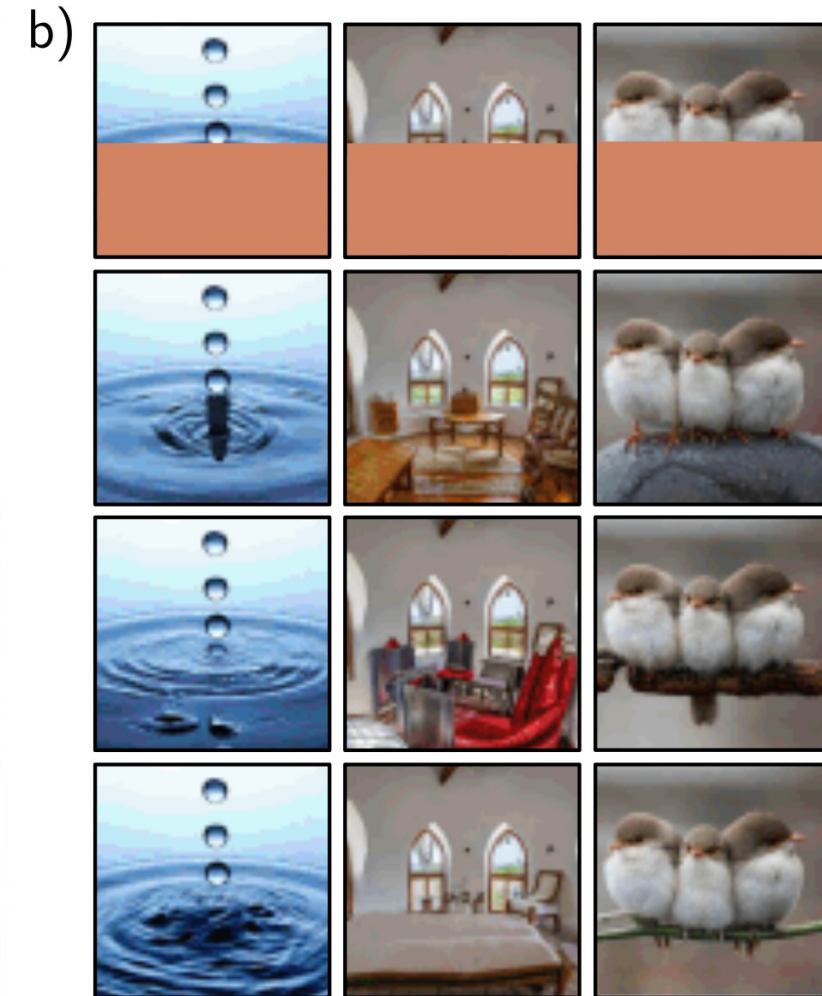
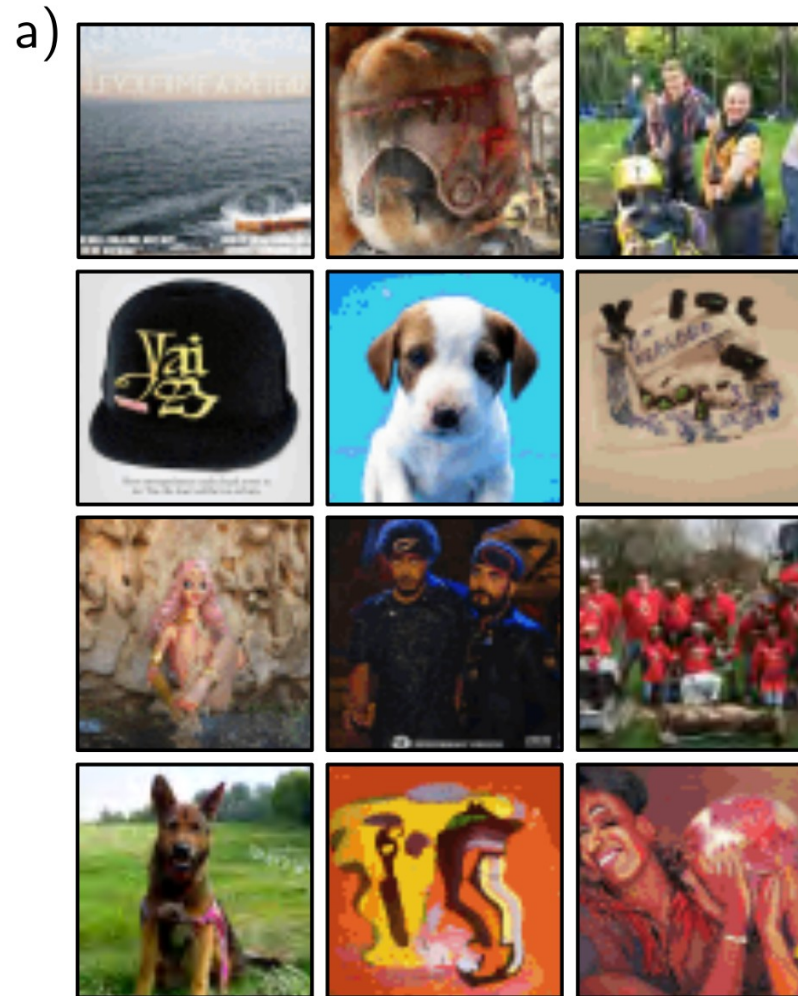
a) Generated from autoregressive ImageGPT model.

Top-left pixel is drawn from estimated empirical distribution,

subsequent pixels are generated.

b) Image completion

Lower half is removed and completed (three versions).



Transformers for Images

- Disadvantage:
 - $O(N^2L)$ complexity of attention
 - e.g. 512 x 512 for language processing
 - For images (e.g. 256 x 256 pixels * 256 x 256 pixels) even more complex
- Solution:
 - More global attention over image patches instead of pixels

Vision Transformer (ViT)

AN IMAGE IS WORTH 16X16 WORDS: TRANSFORMERS FOR IMAGE RECOGNITION AT SCALE

**Alexey Dosovitskiy^{*,†}, Lucas Beyer^{*}, Alexander Kolesnikov^{*}, Dirk Weissenborn^{*},
Xiaohua Zhai^{*}, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer,
Georg Heigold, Sylvain Gelly, Jakob Uszkoreit, Neil Houlsby^{*,†}**

^{*}equal technical contribution, [†]equal advising

Google Research, Brain Team

`{adosovitskiy, neilhoulby}@google.com`

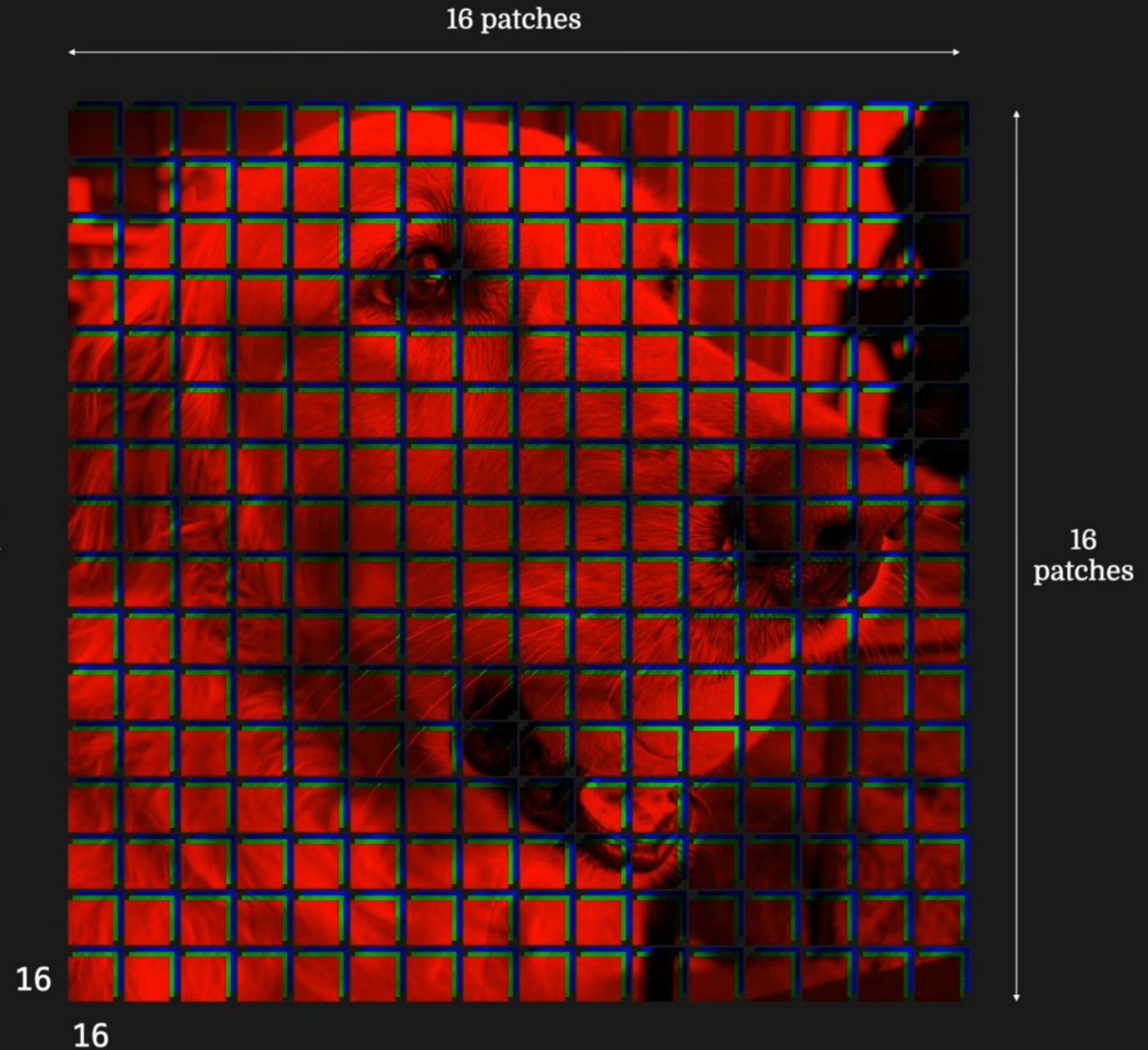
- A pure transformer applied directly to sequences of image patches can perform very well on image classification tasks
- requiring substantially fewer computational resources to train.

Vision Transformer (ViT)

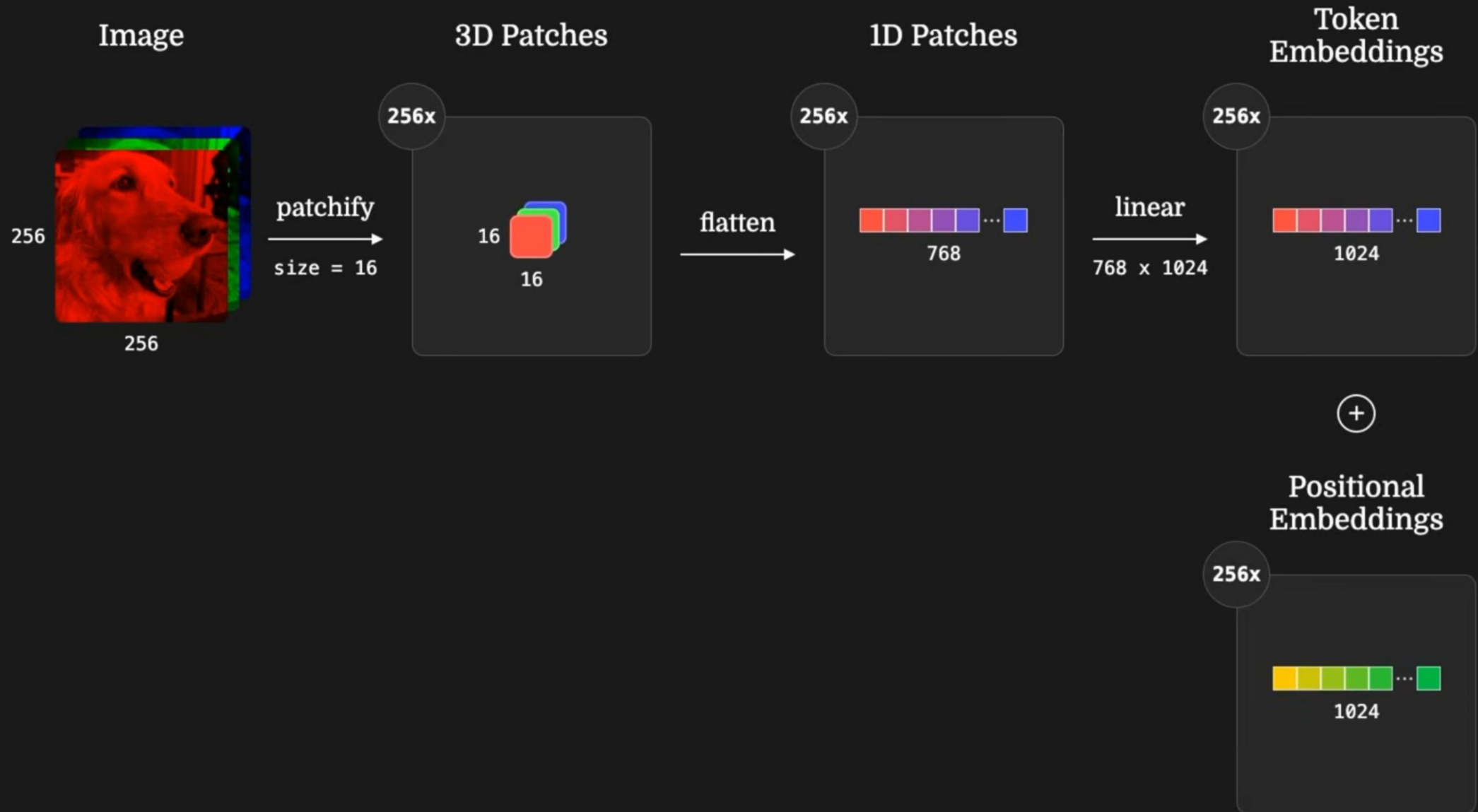
- Tackles the problem of image resolution by dividing the image into 16×16 patches
- Each patch is mapped to a lower dimension via learned linear transformation and fed into the transformer
- Standard learned 1D positional encodings are applied.

Transformers for Images

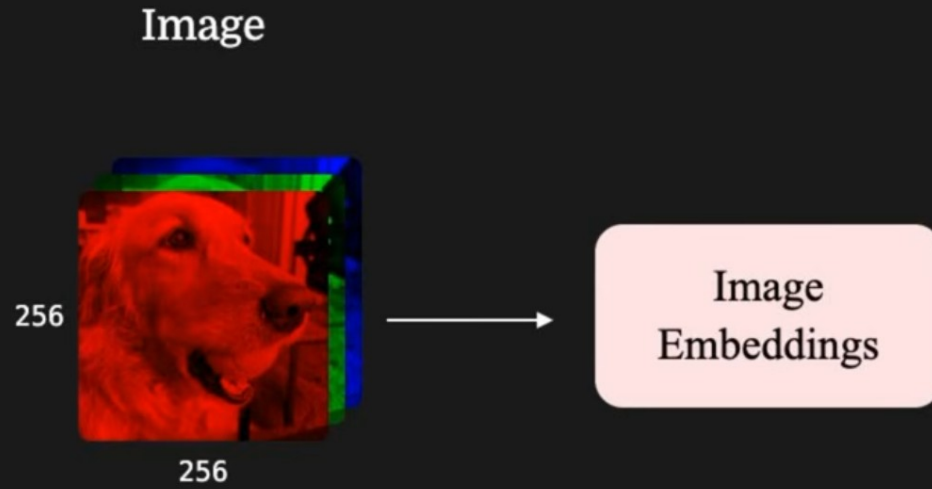
$$\mathcal{O}((16 \cdot 16)^2) \approx 65\text{K}$$



Transformers for Images

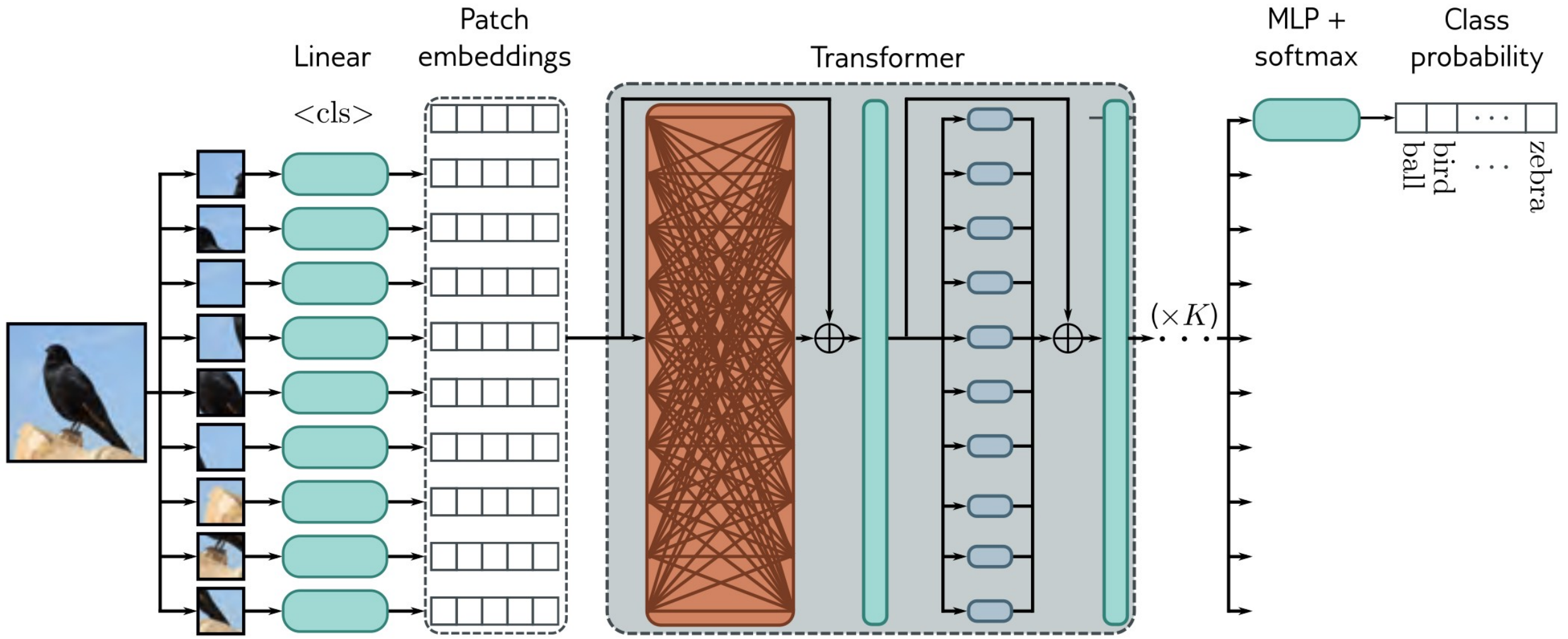


Transformers for Images



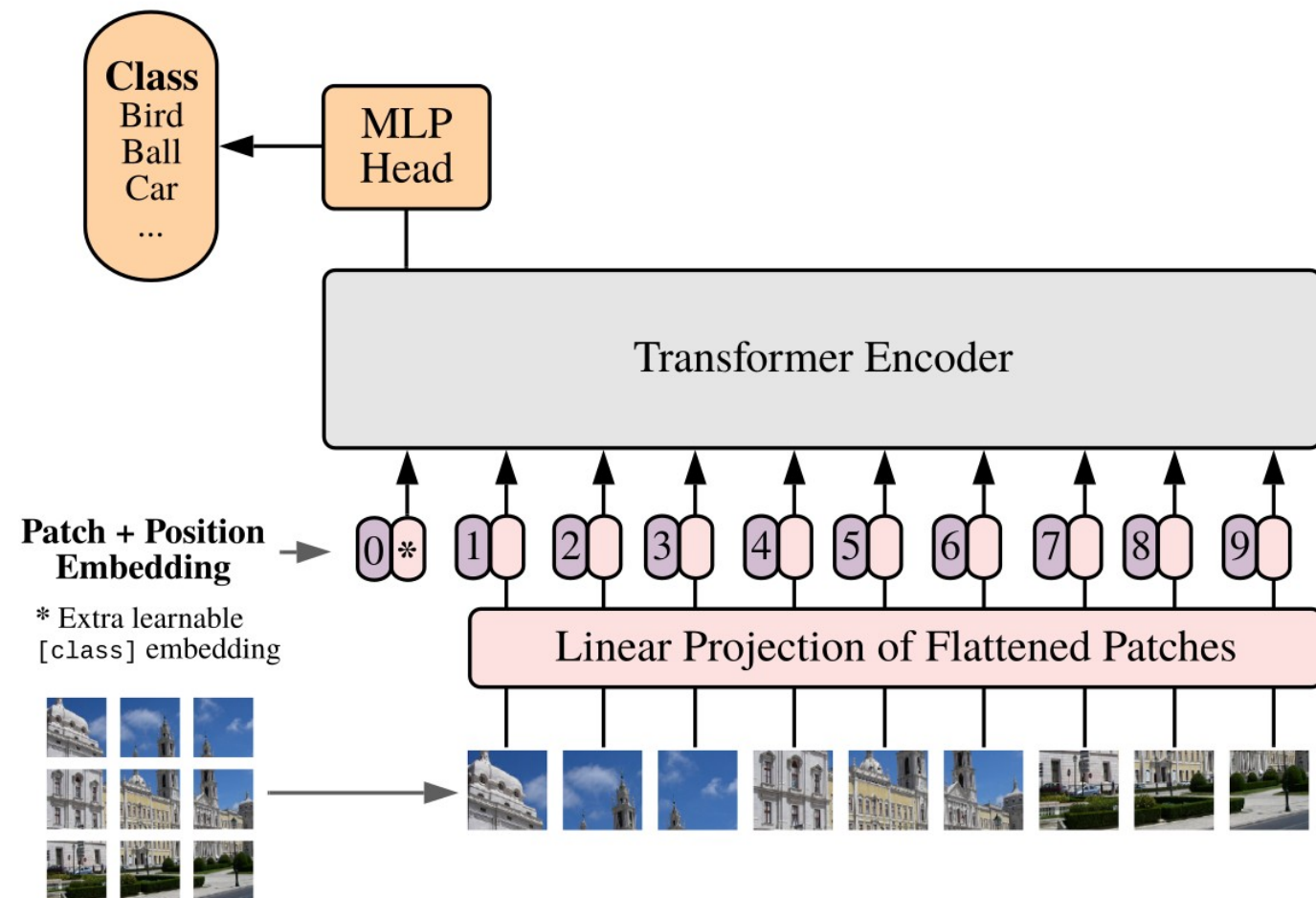
Vision Transformer (ViT)

- Output to the $\langle \text{cls} \rangle$ token is mapped via final network layer and softmax function to class probabilities
- This final layer is fine-tuned to solve custom problems.

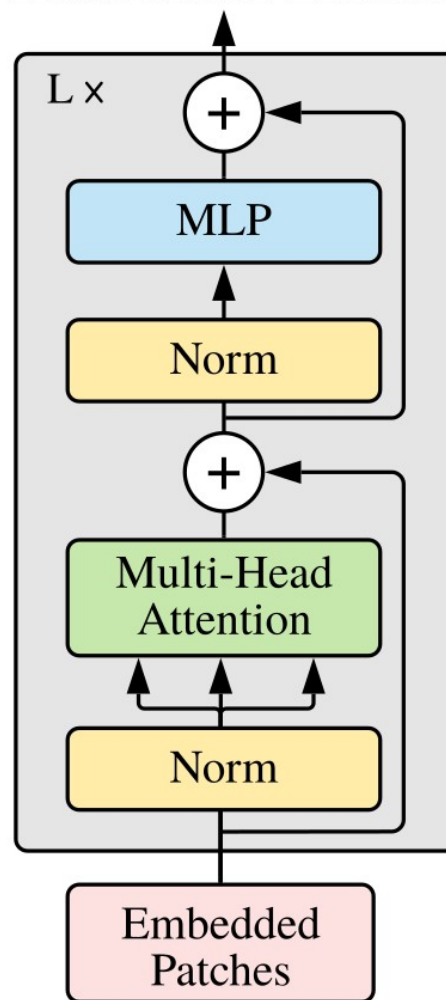


Vision Transformer

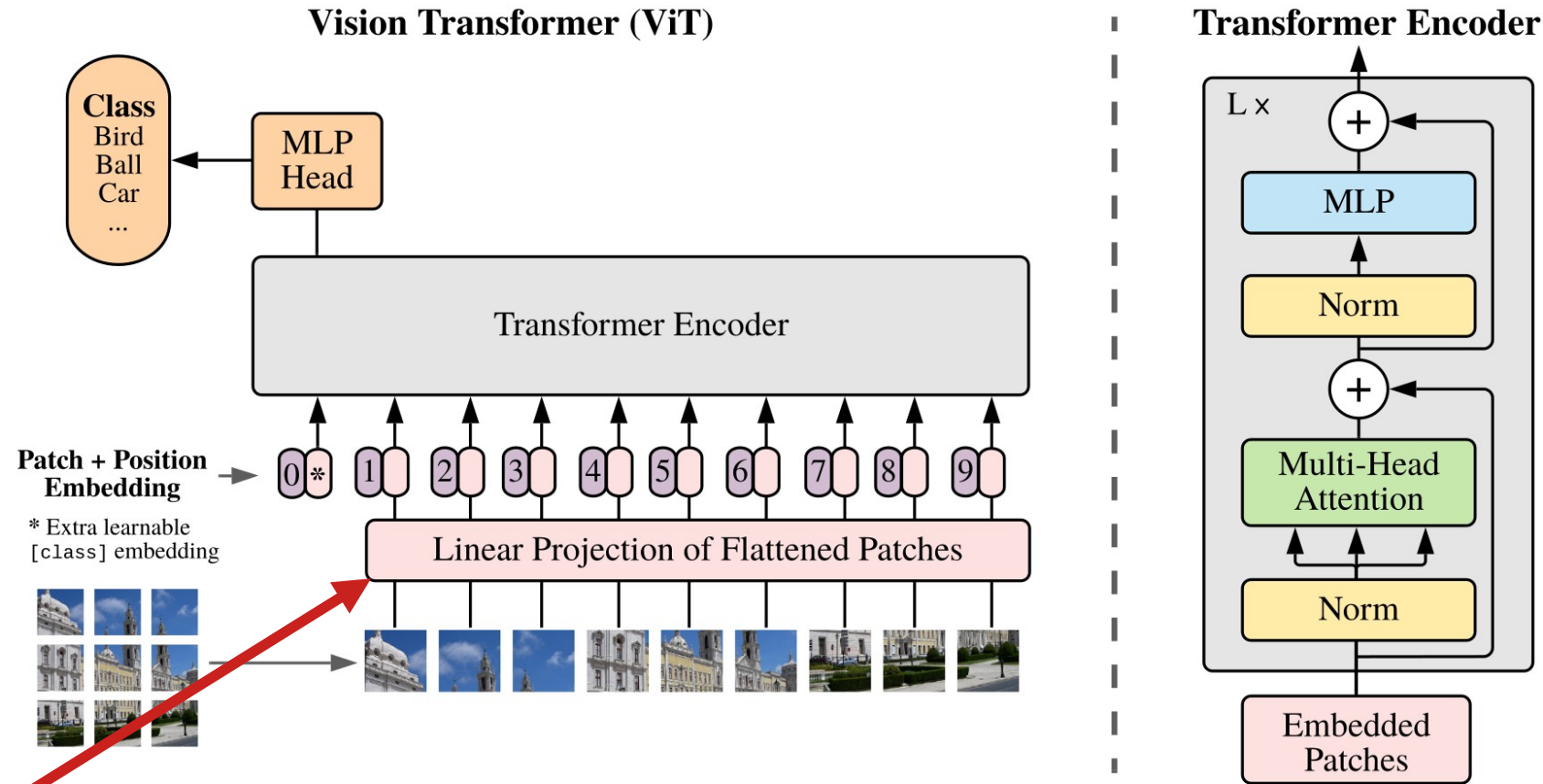
Vision Transformer (ViT)



Transformer Encoder



Vision Transformer

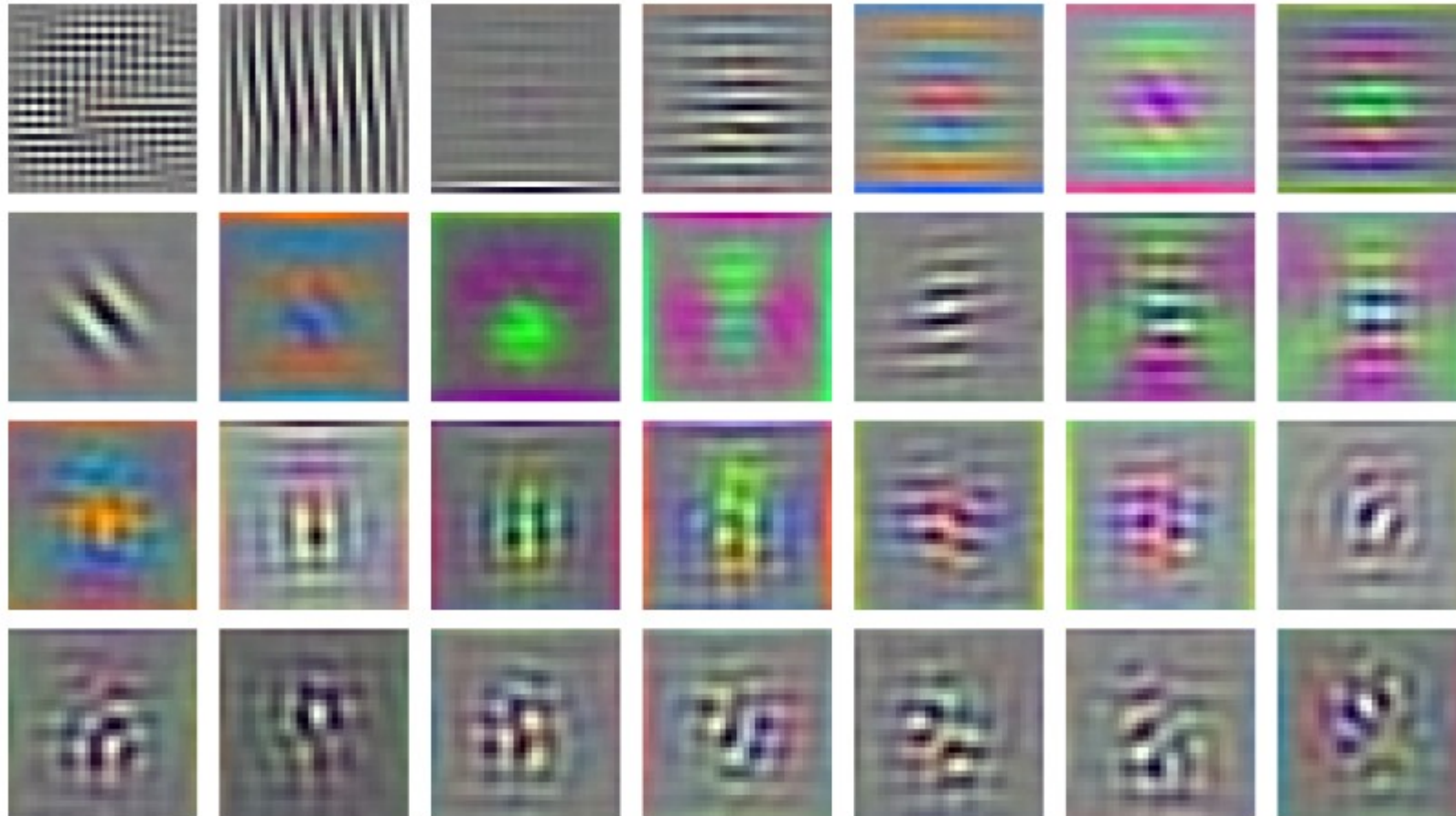


- Linear Projection of Flattened Patches
 - A patch's pixels are stored in flattened vector $\mathbf{f}$
 - Multiplied by embedding matrix $\mathbf{E}$

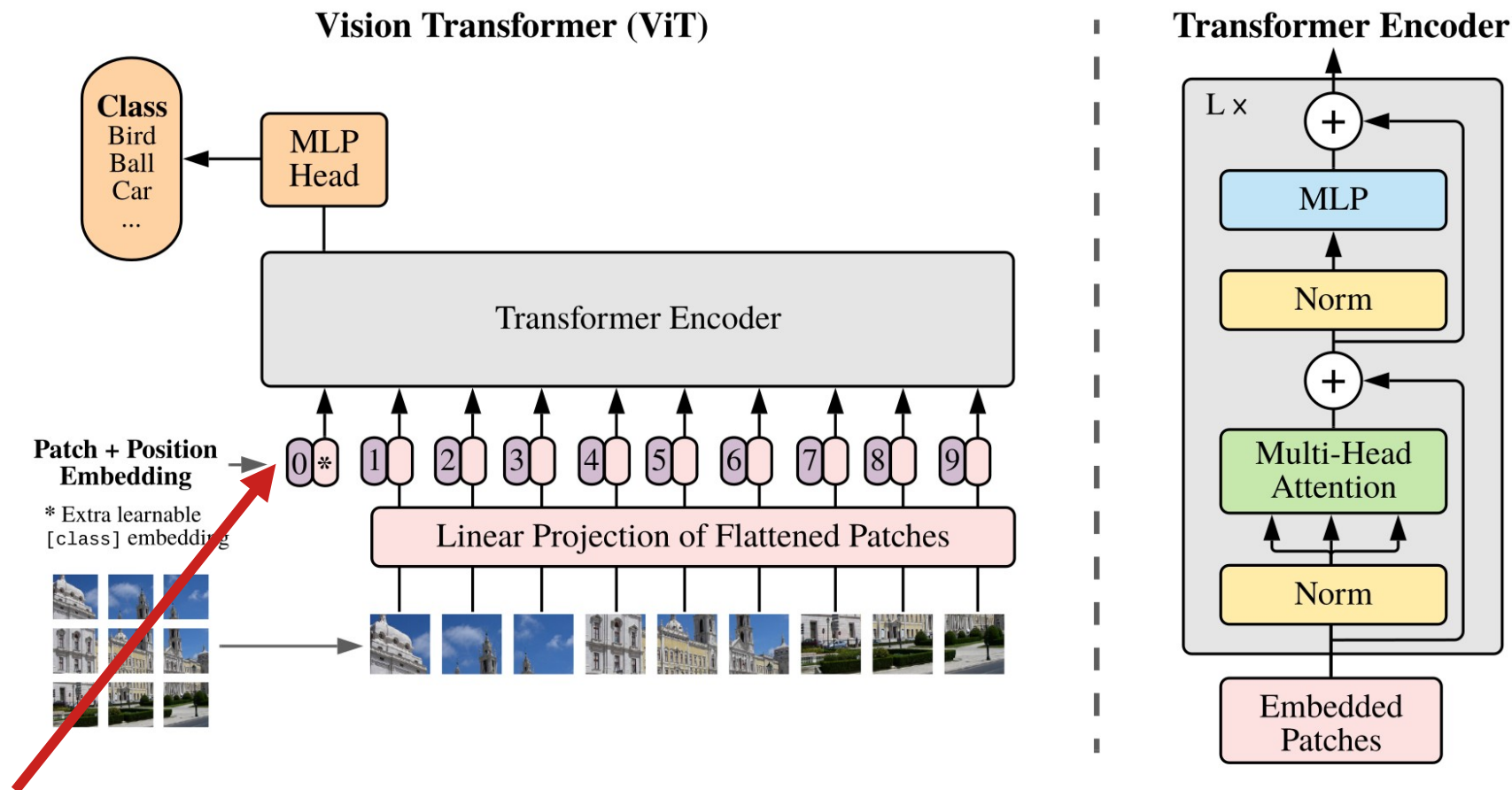
Vision Transformer

- Learned embedding filters

RGB embedding filters
(first 28 principal components)



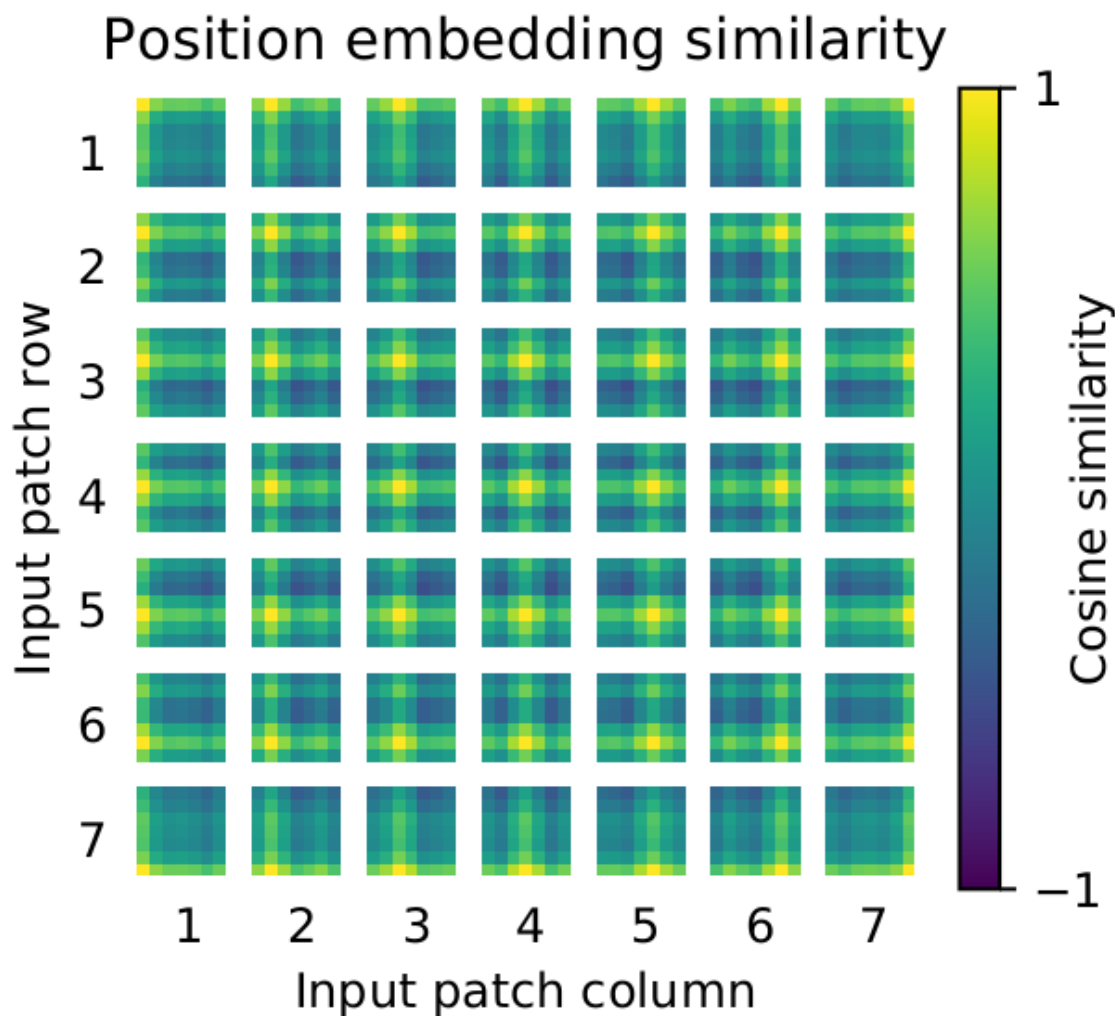
Vision Transformer



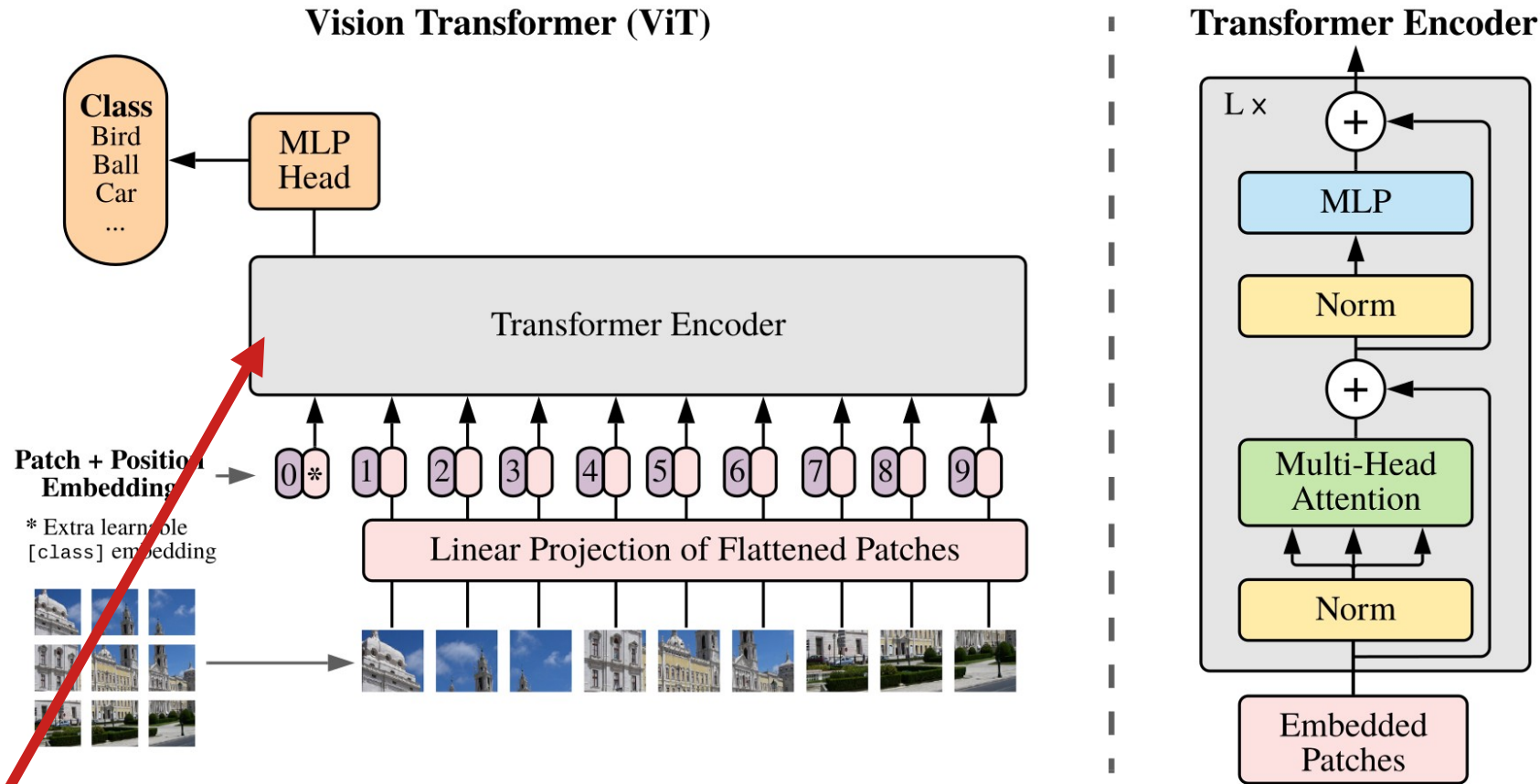
- Positional embedding
 - Is a learned vector for each patch's position

Vision Transformer

- Learned positional embeddings



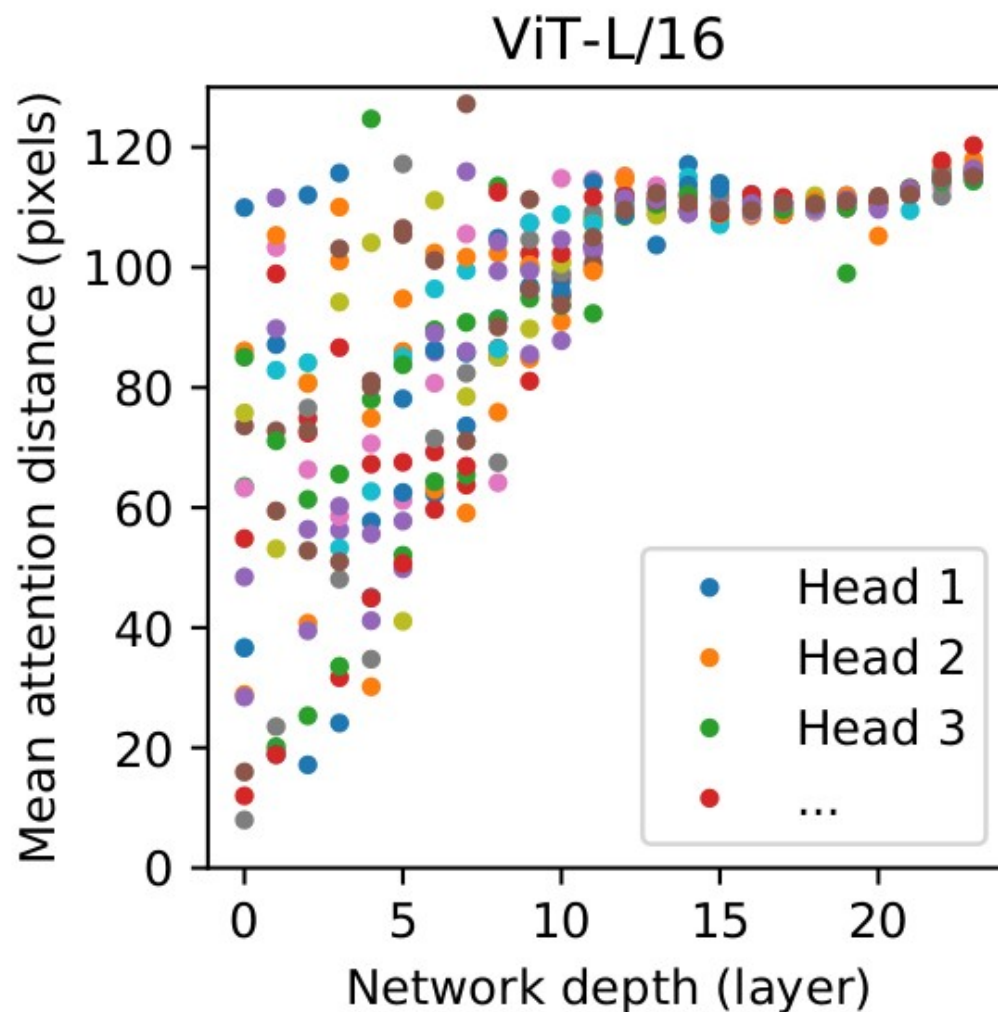
Vision Transformer



- Is a generalization of an MLP
 - as (attention) weights w are computed on the fly

Vision Transformer

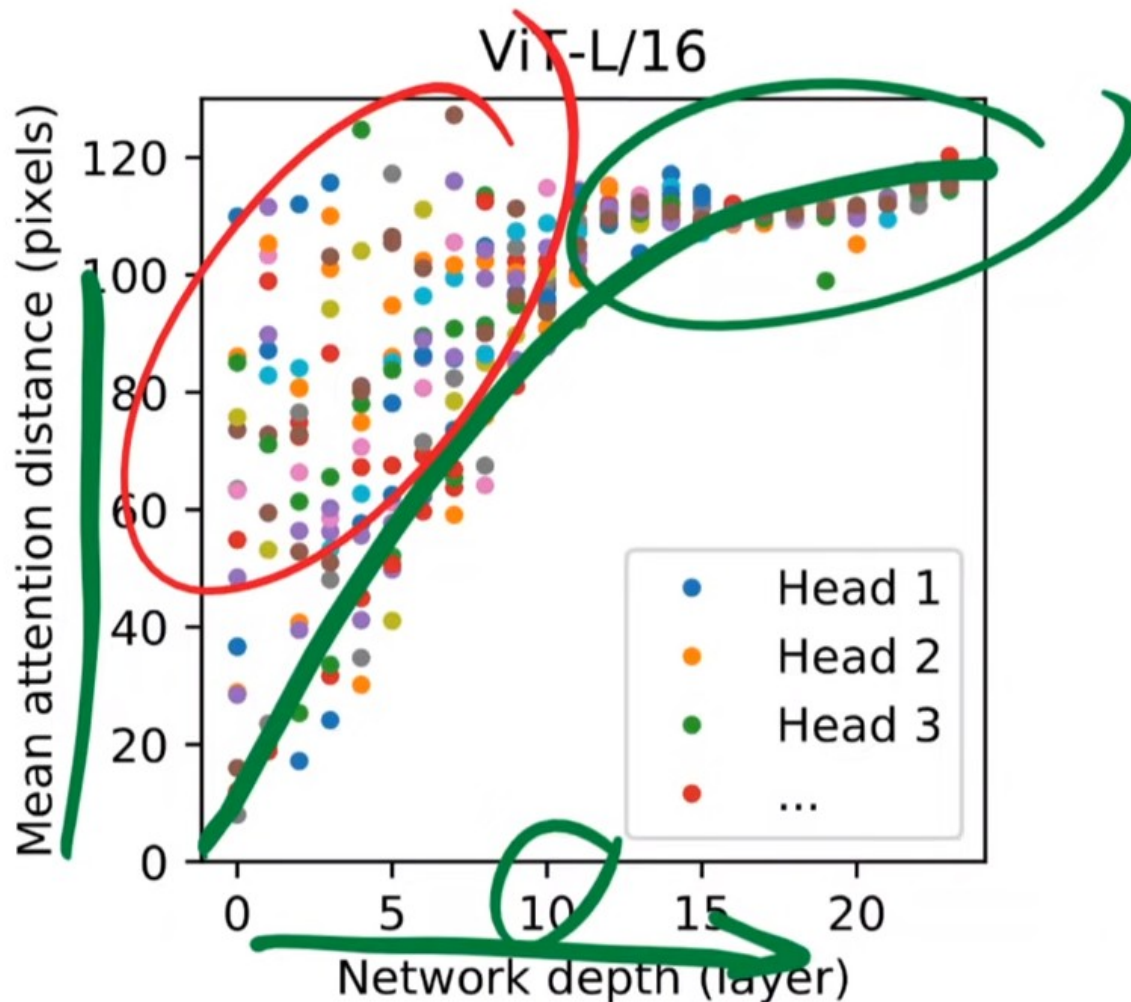
- Mean attention distance



Yannic Kilcher: Review of paper: “An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale (Paper Explained)”.
https://www.youtube.com/watch?v=TrdevFK_am4

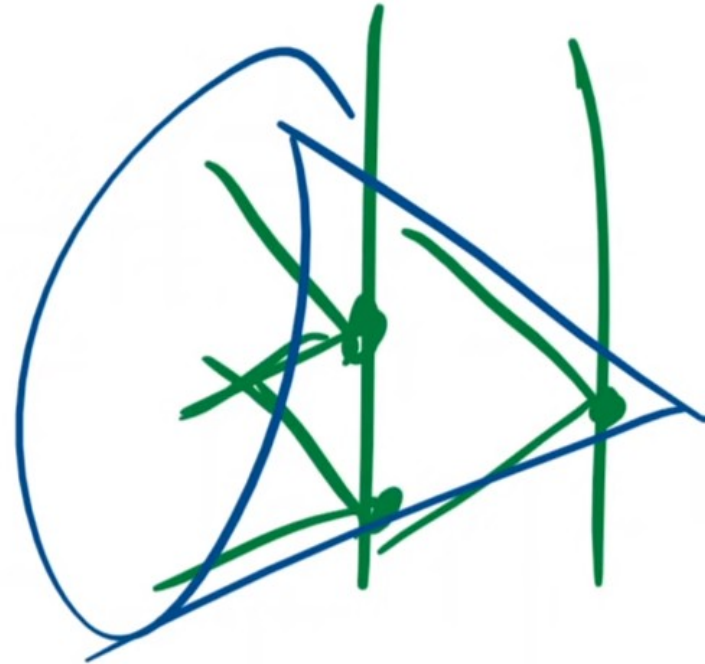
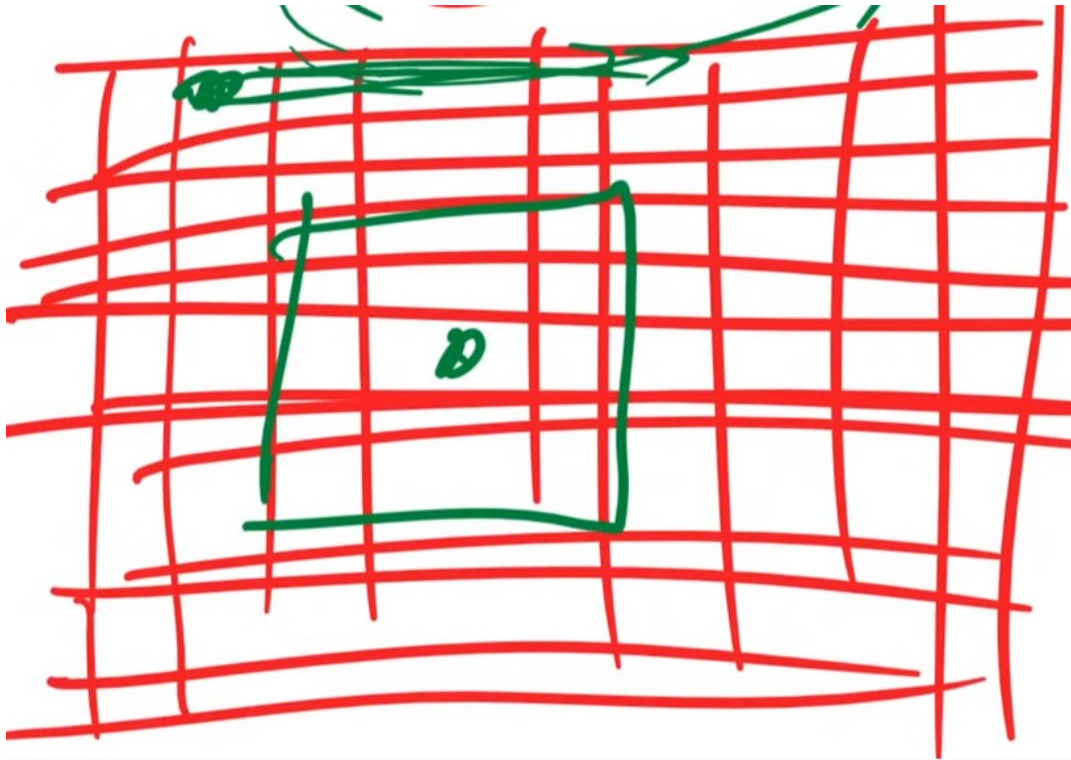
Vision Transformer

- Attention distance of ViT and CNN compared:



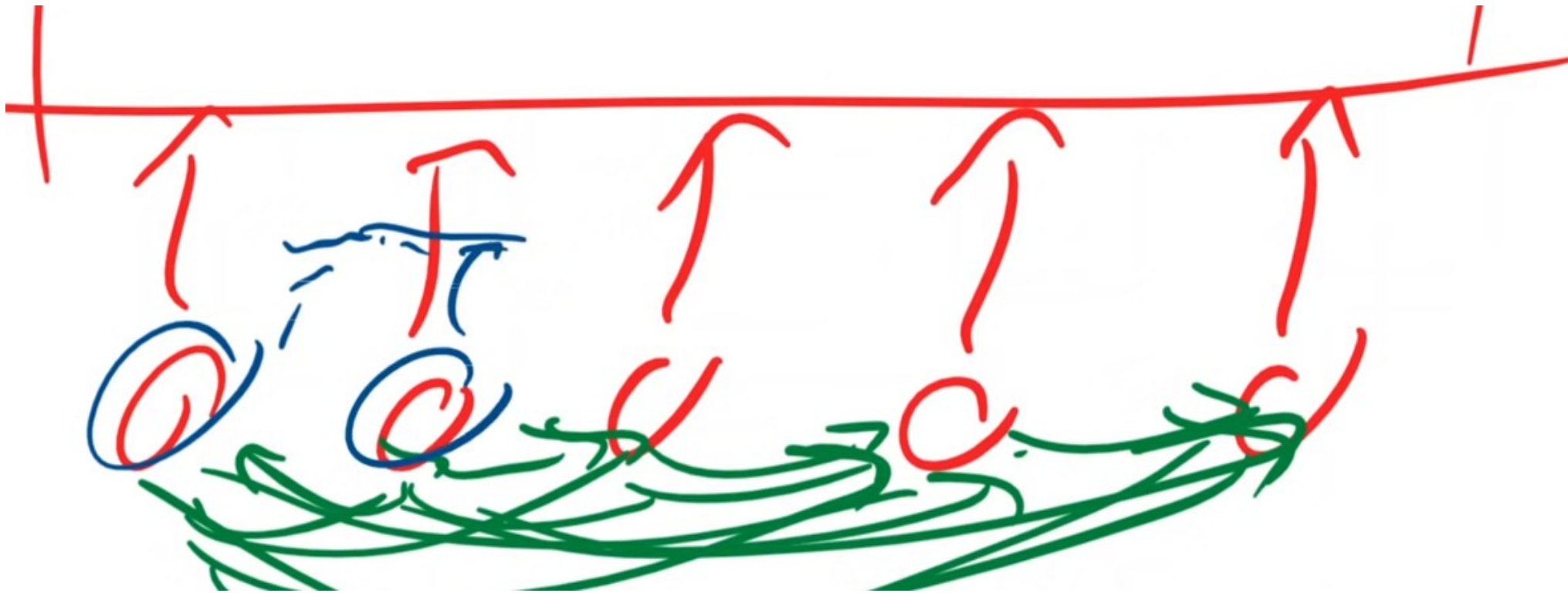
Vision Transformer

- CNN has a good inductive prior/bias
 - A pixels “attends” to its immediate neighborhood.
 - With stacked CNN layers attention is spread out.



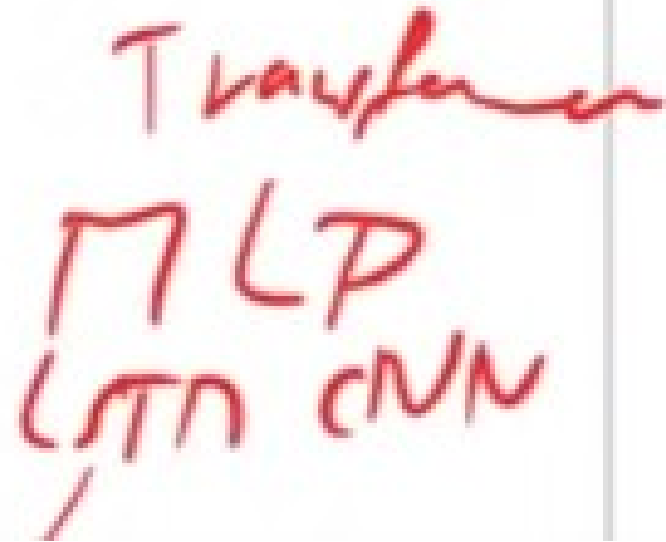
Vision Transformer

- LSTM has a good inductive prior/bias
 - Attention goes through sentence word by word.



Vision Transformer

- Transformers are more general than MLPs.
- LSTMs and CNNs include more inductive biases than MLPs.



Transformer
MLP
LSTM CNN

Vision Transformer (ViT)

- Used supervised pre-training on 303 million labeled images from 18000 classes.
- 11.45 % top-1 error rate on ImageNet benchmark
 - Not as good as the best contemporary convolutional network
 - Strong inductive bias of convolutional networks can only be superseded by employing large amounts of training data.

Multi-Scale Vision Transformers

- ViT operates on single scale (as opposed to CNN)
- Multi-scale transformers
 - Process images at multiple scales
 - Start with high-resolution patches
 - Gradually decrease resolution while increasing number of channels
- Example: **Shifted-Window (SWin) transformer**
 - Patch grid is shifted in adjacent transformers
so that effective receptive field can expand beyond window border.
 - Scale is reduced periodically by concatenating features
from non-overlapping 2x2 patches
Linear transformation maps these features to twice the number of channels.
 - Achieves 9.89% top-1 error rate on ImageNet

Shifted-Window (SWin) Transformer

- Example of multi-scale transformers
- Breaks image into **grid of windows** each processed by transformer
- Each **window subdivided into patches** fed into a transformer.
- **Each alternate layer shifts the window grid** in adjacent transformers
so that **effective receptive** field can expand beyond window border across the whole image.
- Then the **subset of patches interacting** with each other **change**.
- **Scale is reduced periodically** by concatenating features from non-overlapping 2x2 patches
Linear transformation maps these features to twice the number of channels.
- Achieves 9.89% top-1 error rate on ImageNet

Shifted-Window (SWin) Transformer

- a) Original image
- b) Grid of windows
- c) Shifted grid of windows
- d) Larger window with small resolution
- e) Shifted windows at small resolution
- f) Finally, single window

